Mind Uploading and Consciousness Replication

A Typed Ladder for State Capture, Functional Equivalence, Causal Continuity, and Personal Identity

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Abstract

Mind uploading is often discussed as though one operation could be either achieved or not achieved. That framing combines several different scientific, engineering, philosophical, and ethical claims. A measured connectome, runnable circuit model, behaviorally similar agent, autobiographically similar successor, causally continuous process, and phenomenally conscious subject are not the same result. This paper develops a typed evidence ladder separating data capture, structural reconstruction, dynamical reconstruction, held-out functional equivalence, self-model equivalence, causal continuity, phenomenal evidence, and numerical identity. The framework begins with a declared biological state spanning anatomy, molecular and cellular condition, synaptic organization, ongoing dynamics, embodiment, learning history, and self-model. It is motivated by a Self-Aware Networks genealogy extending from a 2011 predictive-network account of self through embodied-copy, multisensory read/write, queryable-network, and multiple-successor proposals. An executable Upload Ladder Benchmark tests embodied recurrent agents under controlled state capture, body transfer, copy branching, perturbation, and matched baselines. Study A compares two bounded 16-node recurrent substitutes across 1,280 bundles and returns zero local-equivalent, five non-equivalent, three unresolved, and one model-disagreement/no-pooling primary disposition. Study B independently implements two learned 32-node substitutes and scores 1,280 opaque held-out jobs before unblinding. Its eight model-specific C6-C7 intervals lie inside the frozen local margin, while 4,131 adverse, 72 null-band, 412 right-censored, and 4,156 failed-intervention records remain visible. The difference between studies is implementation sensitivity, not evidence that uploading succeeded. A warning-free Lean 4 proof leaf checks 22 finite benchmark theorem declarations covering interval disposition, no-pooling, censoring, custody, revocation, rollback, evidence-rung separation, and ordinary-equality fission. A 30-challenge strongest-form review leaves biological state sufficiency, identifiability, architecture-independent replication, long-horizon stability, consciousness evidence, personal identity, and governance open. The paper supplies a falsifiable vocabulary, mathematical accounting system, reproducible synthetic benchmark, adverse evidence, and experimental path. It does not claim biological mind uploading, replicated consciousness, numerical personal identity, or human feasibility.

Keywords: mind uploading; whole-brain emulation; Self-Aware Networks; connectomics; state sufficiency; functional equivalence; causal continuity; consciousness; personal identity; reproducibility

1. Begin with an ordinary problem

Copying a phone is easy to describe. Files, settings, and application state are transferred to another device. If both phones remain active, nobody mistakes them for one physical phone. They begin with similar data and then diverge as soon as they receive different messages, occupy different locations, or are used by different people.

A brain is not a phone. It is a living, changing system whose present operation depends on anatomy, receptors, ion channels, synaptic efficacy, dendritic morphology, glia, neuromodulators, metabolism, recent activity, bodily signals, sensory evidence, learned history, and recurrent interaction with the world. The first scientific problem is therefore not whether a *mind* can be copied. It is to identify which physical and functional variables constitute the target of capture, how accurately each can be measured, and what outcome would count as successful reconstruction.

The second problem is logical. Suppose a new system remembers a person's childhood, speaks in the same voice, solves familiar problems, and sincerely reports being that person. Those observations may support behavioral and autobiographical similarity. They do not by themselves decide whether the original person continued, whether a new person began, or whether the new system has conscious experience. The identity criterion cannot be smuggled into the word *upload*.

The third problem is causal. A static duplicate of one measured instant is not a working life. The reconstructed system must transform new input, sustain internal dynamics, act through a body or interface, receive the consequences of action, learn, and continue through time. The relevant object is a trajectory, not only a snapshot.

Only after these problems are separated is the term **mind uploading** earned. Here it names a research program that attempts to instantiate some declared subset of a person's brain-dependent organization and operation in another substrate. Every reported success must name the subset and the evidence rung reached.

2. The eight claims hidden inside one phrase

This paper separates eight claims.

1. **Data capture:** declared neural and bodily variables were measured at stated resolution and time.
2. **Structural reconstruction:** declared organization was reproduced within measured error.
3. **Dynamical reconstruction:** the new system reproduced relevant state transitions, not merely a frozen graph.
4. **Functional equivalence:** held-out perception, memory, skill, choice, adaptation, and action tests were matched.
5. **Self-model equivalence:** the system represented its body, history, capacities, limitations, goals, and authorship in comparable ways.
6. **Causal continuity:** a declared continuous or staged process linked the source and successor.
7. **Phenomenal continuity:** evidence relevant to conscious experience was obtained under an explicit theory and test.
8. **Personal or numerical identity:** a philosophical and legal criterion for being the same person was stated and defended.

These rungs are related, but none should inherit the name of a higher rung merely by success at a lower rung. A complete connectome is a major structural accomplishment. A connectome-constrained model that predicts a sensorimotor pathway is a major functional accomplishment. Neither result alone establishes a conscious continuation of an individual.

3. What contemporary neuroscience already establishes

3.1 Wiring diagrams are powerful but typed

The complete *C. elegans* wiring diagram established that an entire nervous system could be reconstructed at cellular and synaptic scale [1]. Later connectomes revealed sex differences and developmental rewiring [2,3]. The 2024 adult *Drosophila* connectome expanded the scale to roughly 140,000 neurons and tens of millions of synapses, with cell annotations, predicted neurotransmitter identities, network motifs, and region structure [4-6].

These maps support real computations. Shiu and colleagues constructed a whole-fly connectome model that predicted coarse sensorimotor transformations and generated experimentally testable circuit hypotheses [7]. Connectome-constrained models can predict neural activity in the fly visual system when connection strengths and task constraints are incorporated [12]. These achievements show that topology contains substantial causal structure.

They also show why the result must be typed. Model predictions depend on assumed signs, weights, cell rules, inputs, and thresholds. In *C. elegans*, direct optogenetic perturbation revealed signal propagation and extrasynaptic effects not available from anatomy alone [8]. Combining a connectome with whole-brain perturbation recordings constrained causal dynamics more strongly than the graph by itself [9]. Connectome-constrained recurrent networks can share wiring and weights while producing different dynamics under uncertain biophysical parameters; partial activity recordings can reduce that degeneracy [10].

The strongest scientific account is therefore constructive: anatomy supplies a powerful constraint; molecular identity, physiology, dynamics, and perturbation supply additional constraints; embodiment tests whether the resulting controller works in a closed loop.

3.2 A cell is more than an edge in a graph

Patch-seq joins electrophysiology, gene expression, receptors, and channels in the same cell [13]. Integrated morphoelectric and transcriptomic studies show both stable classes and meaningful within-class variation [14-17]. Complete neuronal morphology adds information about input, transformation, projection, and circuit position [16]. Direct comparisons of anatomy and physiology show that functional weights and subcellular computation require physiological parameterization [18].

This matters because a reconstruction target cannot be declared complete merely because every cell and synapse has a coordinate. A scientifically defensible target must state which variables it omits and measure the behavioral or dynamical consequence of each omission.

3.3 Memory is constrained by structure but instantiated dynamically

Optogenetic experiments show that manipulating memory-linked ensembles can alter recall [19]. Long-term associative memory involves evolving ensemble dynamics [20]. Stable behavior can coexist with changing population activity, and representational drift can occur even for familiar stimuli [21,22]. A successful reconstruction may therefore preserve content or capability without reproducing every original carrier, while another reconstruction may preserve a graph but fail to reproduce the relevant dynamics.

3.4 Body and return pathways matter

Body ownership can be shifted by coordinated multisensory evidence [23]. Intracortical interfaces can support robotic reach and grasp [24], and tactile cortical feedback can improve robotic control [25]. Brain-machine performance emerges through interaction between decoder design and neural adaptation [26]. An upload experiment that changes body, sensory latency, action space, or feedback statistics is not merely moving the same controller to a neutral container. It is creating a new closed-loop condition that can change the controller.

4. The SAN genealogy: from distributed self to a typed upload program

The historical development matters because later terminology should not be projected backward.

In 2011, Blumberg described the self as a prediction produced by a distributed neural network rather than a hidden observer. In 2012, an embodied-self discussion treated memories, connections, nervous system, organs, ecosystem, and artificial-body remapping as different contributors to continuity. The same period's Neo Mind Cycle work demonstrated a reciprocal brain-computer feedback loop, an ancestor of later external-cortex proposals.

Two 2013 discussions supply the core identity argument. In one, a proposed copy is compared with a changing ocean: measurement, reconstruction, and environmental divergence occur while the target itself changes. In another, an initially similar brain and body immediately occupy a different location, receive different evidence, and begin a different action-consequence history. These are not proofs that reconstruction is impossible. They are early formulations of a situated-trajectory test.

The 2017 Neural Lace program made the engineering ambition explicit. It proposed multisensory reading and writing, current-state estimation before intervention, and transmission of the difference between the present and target pattern. *The Brain as a Special Kind of Hard Drive* discussed uploading and downloading parts of selves, memory sharing, and modification while also describing the brain as a massively parallel, state-dependent system. "Hard drive" was an engineering provocation, not a claim that living neural state is a static disk image.

In 2021, one source described a computer becoming part of a larger mind through reciprocal feedback. The original audio was recorded on 11 October 2021 at 07:08:05 UTC; from 7:54.44 to 8:31.04 it describes EEG-derived light and sound returning to the participant and being modeled again by the brain, and from 8:56.18 to 9:22.72 it proposes extending the mind into a larger computer-supported mind [41]. A second recording, created the same day at 03:34:02 UTC, proposes querying a running brain's data structure instead of copying every detail; its complete spoken claim occupies 0:00-0:42.97

[42]. The latter recording's later written TMS addendum is not part of that 43-second spoken source and is not used to enlarge the 2021 claim. In 2022, the program explicitly considered backups, multiple successors, customization, and ethical risk. Mature SAN work added a distinction between longer-lived structural constraints and fast recurrent activity. By 2024, the book architecture separated upload, download, customization, and the problem of moving from a rendered perspective to a causally active conscious perspective.

The 2017 article and podcast also require separate attribution. The article is an authored synthesis with its own public dateline. The podcast contains a recorded introduction and a multi-speaker interview. A paragraph-level concordance identifies which article claims are direct transcript matches, which are related live-discussion material, and which are article-only synthesis [43]. The explicit uploading, downloading, memory-sharing, and modification sentence is supported by the public article itself; it is not represented as a verbatim Android Jones interview statement.

The present paper joins those stages into a typed ladder. It does not claim that the exact ladder existed in 2011, 2012, or 2017.

5. A declared state model

Let the source organism's state at time (t) be

$$X(t) = \{A, M, C, W, D, B, H, S, E\}_t, \qquad [1]$$

where A is anatomy and connectivity, M molecular state, C cellular physiology, W synaptic efficacy and plasticity variables, D ongoing neural dynamics, B body and interoceptive state, H learning and causal history, S self-model state, and E current environment and relationships. This is a bookkeeping set, not a claim that every member is equally necessary.

A measurement process observes only part of that state:

$$Y = \mathcal{M}_{q,\tau}(X) + \varepsilon, \qquad [2]$$

where q records modalities and resolution, τ the acquisition interval, and ε measurement error. A reconstruction operator produces an initial successor state

$$\widetilde{X}(0) = \mathcal{R}_\phi(Y, \Pi), \qquad [3]$$

with implementation parameters ϕ and declared prior assumptions Π.

The biological and reconstructed systems then evolve:

$$X(t+1) = F(X(t), u(t), B(t), E(t)), \qquad [4]$$

$$\widetilde{X}(t+1) = \widetilde{F}(\widetilde{X}(t), \widetilde{u}(t), \widetilde{B}(t), \widetilde{E}(t)). \qquad [5]$$

Equations [4] and [5] make embodiment explicit. Even identical initial internal states can diverge under different inputs, bodies, and environments.

For each declared state family (k), normalized capture error is

$$e_k = \frac{d_k(P_k X, P_k \widetilde{X})}{Z_k}, \qquad [6]$$

where P_k projects the state onto the measurable family, d_k is a declared distance, and Z_k is a scale factor. State coverage is

$$C_{state} = \frac{\sum_k w_k I_k (1 - e_k)}{\sum_k w_k}, \qquad [7]$$

where $I_k = 1$ only when family k was actually measured. Unmeasured variables cannot receive zero error by assumption.

6. From state similarity to functional equivalence

Structural similarity is tested with anatomy and cell-state metrics:

$$E_{str} = \alpha_A e_A + \alpha_M e_M + \alpha_C e_C + \alpha_W e_W. \qquad [8]$$

Dynamical similarity is evaluated across held-out inputs and initial states:

$$E_{dyn} = \mathbb{E}_{u,x_0}\left[d_D(T_F(x_0, u), T_{\widetilde{F}}(\widetilde{x}_0, u)) \right], \qquad [9]$$

where T denotes a trajectory. A model that reproduces spontaneous activity but fails under perturbation has not passed the same test as one that reproduces causal response.

Functional equivalence is task-indexed:

$$E_{fun} = \sum_{j=1}^{J} \beta_j \, d_j(o_j, \widetilde{o}_j), \qquad [10]$$

for held-out perception, memory, skill, choice, planning, adaptation, and action outcomes o_j. The task weights β_j must be frozen before evaluation.

Causal equivalence requires matched intervention response:

$$E_{causal} = \mathbb{E}_{a \in \mathcal{A}}\left[d\left(P(O \mid do(a)), P(\widetilde{O} \mid do(a)) \right) \right]. \qquad [11]$$

This is stronger than passive correlation. The intervention set should include perturbation, damage, rescue, sensory conflict, body change, and novel goals.

A self-model test compares declared internal estimates:

$$E_{self} = d_S\left(S_{source}, S_{successor}\right) + \lambda \, E_{calibration}, \qquad [12]$$

where calibration measures whether confidence about body, memory, authorship, limitations, and uncertainty matches performance.

No single weighted average should erase the profile. The result is reported as a vector:

$$\mathbf{L} = (C_{data}, C_{str}, C_{dyn}, C_{fun}, C_{self}, C_{causal}, C_{phen}, C_{id}) . \qquad [13]$$

The vector is the **Upload Evidence Ladder**. A project may report, for example, strong structural and moderate functional scores while leaving phenomenal and identity claims unassigned.

7. The person is a trajectory, not a checksum

Let a person's realized trajectory be

$$\Gamma = [X(t_0), a(t_0), r(t_0), \ldots, X(t_n)], \qquad [14]$$

where actions (a) produce returned consequences (r). Two successors initialized from similar reconstructions diverge according to

$$\Delta_{12}(t) = d\left(\widetilde{X}_1(t), \widetilde{X}_2(t)\right). \qquad [15]$$

If their bodies, inputs, random events, or choices differ, then generally

$$\Delta_{12}(t+1) > 0 \quad \text{for some } t \geq 0. \qquad [16]$$

This formalizes the 2013 situated-copy argument. Immediate divergence does not erase shared ancestry, memory, or similarity. It prevents those properties from being silently equated with one continuing numerical individual.

Numerical identity behaves like equality. If original (O) is numerically identical to successor (A), and (O) is numerically identical to successor (B), transitivity implies

$$O = A \wedge O = B \Rightarrow A = B. \qquad [17]$$

When A and B are two independently accessible physical instances, $A \neq B$. A branching procedure therefore cannot assign simple numerical identity from one source to both successors without changing the identity concept. It may instead assign ancestry, psychological continuity, legal succession, or another explicitly typed relation. This is a logical constraint, not a test for consciousness.

Causal continuity can be represented as a directed provenance graph (G_c=(V,E_c)). A successor continuity score may use

$$C_{causal} = \frac{\sum_{e \in P(O,S)} w_e \, v_e}{\sum_{e \in P(O,S)} w_e}, \qquad [18]$$

where $P(O,S)$ is the declared path from source to successor and v_e verifies each preservation, replacement, coadaptation, or transfer event. The score measures documented process continuity. It does not settle whether that process is metaphysically sufficient for being the same person.

8. What SAN adds

SAN treats the self as a distributed, embodied, changing operation rather than a record inspected by an inner observer. Longer-lived structural and molecular constraints shape which patterns can be received and reinstated. Fast neural dynamics provide the present context. Recurrent traffic joins partial sensory, memory, body, and action information. A self-model tracks the system's body, capacities, history, and current control. Action changes the world and returns new evidence.

For reconstruction, this yields a concrete dependency chain:

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memory-bearing structure and cellular state
-> current receiver readiness and ongoing dynamics
-> recurrent self/world rendering
-> choice and embodied action
-> returned consequence
-> learning, drift, and updated self-model
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The chain explains why a static connectome is a valuable but incomplete capture target. It also explains why exact microscopic duplication may not be necessary for every functional success: relational patterns can sometimes survive carrier change. The experimental burden is to discover which invariants preserve which functions under which substitutions.

SAN's receiver-relative view also changes the upload metaphor. A running brain may be interrogated through perturbation and response rather than copied only as a frozen object. Let a query (q__i) produce response (z__i):

$$z_i = \mathcal{Q}(X, q_i) + \eta_i. \qquad [19]$$

An adaptive query policy chooses the next intervention from remaining uncertainty:

$$q_{i+1} = \arg\max_q \mathbb{E}\left[\mathcal{I}(X; z \mid q, \mathcal{D}_i) - \kappa(q)\right], \qquad [20]$$

where $\mathcal{I}$ is expected information gain and κ is biological, ethical, and measurement cost. This turns the 2021 "query the brain's data structure" idea into a bounded system-identification program. It does not assume that queries can recover every hidden state.

9. The Upload Ladder Benchmark

The first executable application should use artificial embodied agents because the complete source state is known. A recurrent source agent learns navigation, object recognition, autobiographical events, preferences, tool use, damage recovery, and a body schema in a seeded environment. The system then creates successors from controlled capture packages.

9.1 Capture conditions

1. output behavior only;
2. policy parameters only;
3. policy plus recurrent hidden state;

4. policy, hidden state, and episodic memory;
5. full controller graph plus optimizer and plasticity state;
6. full controller plus calibrated body model;
7. full state with staged coadaptive replacement;
8. full state followed by two-copy branching.

The capture budget is

$$B_{cap} = B_A + B_M + B_C + B_W + B_D + B_B + B_H + B_S. \qquad [21]$$

Each condition declares which terms it includes. Capacity- and compute-matched controls prevent a larger successor from winning merely through extra resources.

9.2 Held-out tests

Successors face novel mazes, unfamiliar objects, delayed autobiographical queries, altered bodies, sensory latency, conflicting self-cues, goal reversal, damage, repair, and social recognition. A blinded evaluator compares source and successors. Internal tests compare dynamics, causal interventions, memory provenance, self-model calibration, and adaptation.

The benchmark score remains vector-valued:

$$\mathbf{B} = (1 - E_{str}, 1 - E_{dyn}, 1 - E_{fun}, 1 - E_{causal}, 1 - E_{self}, R_{robust}, R_{transfer}). \qquad [22]$$

A scalar summary may be reported only as a secondary convenience:

$$B_{min} = \min_j B_j, \qquad [23]$$

because the minimum exposes a missing load-bearing dimension that an average could hide.

9.3 Branching and report tests

After one capture, successors (A) and (B) receive different experiences. The benchmark records shared source memory, later divergence, conflicting ownership claims, and their ability to distinguish inherited from newly acquired memories. Reports of “I am the original” are treated as self-model outputs to be tested for calibration, not as automatic identity verdicts.

The fuller state account earns its additional complexity only through held-out incremental value. Let M_0 be a connectome-only model and M_1 the fuller state-and-trajectory model:

$$\Delta V = V(M_1) - V(M_0). \qquad [24]$$

If ΔV is indistinguishable from zero under adequate power, matched compute, and relevant perturbations, the added variables have not earned their complexity for that endpoint. This narrows the application of the fuller model; it does not turn a local result into a global claim about SAN.

9.4 Normalization, equivalence, and uncertainty

Every score must be operational before the first controlled run. For a lower-is-better loss $\ell_{m,r}$ on metric m and run r, let ℓ_m^* be the frozen source-to-source or measurement-floor loss and ℓ_m^0 the frozen null-baseline loss. The normalized error is

$$e_{m,r} = \mathrm{clip}_{[0,1]}\left(\frac{\ell_{m,r} - \ell_m^*}{\ell_m^0 - \ell_m^*}\right), \qquad \ell_m^0 > \ell_m^*. \qquad [25]$$

For a higher-is-better quantity $q_{m,r}$, the corresponding normalized score is

$$s_{m,r} = \mathrm{clip}_{[0,1]}\left(\frac{q_{m,r} - q_m^0}{q_m^* - q_m^0}\right), \qquad q_m^* > q_m^0. \qquad [26]$$

If either anchor collapses, the metric is not identified and must be reported as unresolved rather than assigned an arbitrary zero or one. Missing state families remain missing.

Functional equivalence is indexed to a frozen distribution of held-out tasks. For tolerance vector $\boldsymbol{\epsilon}$ and failure probability δ, define

$$X \sim_{\mathcal{T},\boldsymbol{\epsilon},\delta} \widetilde{X} \iff \Pr_{\tau\sim\mathcal{T}}\left[\bigcap_m d_m(X, \widetilde{X} \mid \tau) \le \epsilon_m\right] \ge 1 - \delta. \qquad [27]$$

This is an endpoint-specific equivalence relation only when the declared distances and tolerances satisfy the required reflexive, symmetric, and transitive conditions. Otherwise it is an acceptance rule, and the report must use that weaker name.

Causal-graph recovery and intervention-response recovery remain distinct:

$$S_{graph} = F_1(E, \widetilde{E}), \qquad E_{do} = \frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}} D\Big(P(O \mid do(a)), P(\widetilde{O} \mid do(a))\Big). \qquad [28]$$

A matching edge set can coexist with the wrong signs, strengths, delays, or nonlinear responses. Passing S_{graph} therefore cannot substitute for passing E_{do}.

When two systems execute the same operation at different speeds, trajectory comparison may allow a monotone time map γ:

$$d_{warp}(z, \widetilde{z}) = \inf_{\gamma\in\Gamma}\left[\frac{1}{T}\int_0^T d(z(t), \widetilde{z}(\gamma(t)))\, dt + \lambda_\gamma\, \Omega(\gamma)\right]. \qquad [29]$$

The penalty Ω prevents unconstrained warping from hiding altered causal order, latency, or missing events. Both warped and unwarped distances must be disclosed.

Uncertainty is estimated by resampling source agents as the outer cluster and reconstruction seeds within source-condition pairs as the inner cluster. For metric m, the report provides the point estimate, 95% cluster-bootstrap interval, source-to-source variance, reconstruction-seed variance, and the complete per-run values. Equivalence is declared only when the confidence interval lies inside the preregistered equivalence margin; failure to reject a difference is not equivalence.

The minimum-dimension score in Equation [23] is interpretable only when every included component has a validated direction, noncollapsed anchors, a common $[0,1]$ range, and a reported uncertainty interval. If any load-bearing component is missing or unresolved, B_{min} is unassigned and the full vector remains primary.

9.5 Frozen executable schema

The Source-and-Citation Stage benchmark schema binds each run to a source seed, source checkpoint hash, capture condition, reconstruction seed, body condition, held-out world split, intervention schedule, evaluator-blinding key, software version, and result-bundle hash [44]. It also binds every metric to direction, units, normalization anchors, aggregation level, uncertainty estimator, equivalence margin, and missing-data rule. A conforming result cannot use a field outside the schema without recording a versioned schema amendment.

Adaptive queries use an ensemble or posterior over candidate source models. The next query is selected by estimated entropy reduction minus registered biological, ethical, and measurement cost:

$$\widehat{q}_{i+1} = \arg\max_{q\in\mathcal{Q}_{safe}} \Big[H(\widehat{p}_i(X)) - \mathbb{E}_{z\sim\widehat{p}(z|q,\mathcal{D}_i)} H(\widehat{p}_{i+1}(X \mid z,q)) - \kappa(q)\Big]. \qquad [30]$$

The estimator must be calibrated on synthetic systems with known hidden state before it can support a biological acquisition claim.

Memory provenance is reported with classwise precision and recall over inherited, acquired, inserted, and unresolved events. Self-model confidence uses a preregistered proper score [50] and expected calibration error. Branch divergence time is the first time at which a registered distance exceeds threshold θ for K consecutive windows:

$$t_{div} = \inf\{t : \Delta_{12}(t : t+K-1) > \theta\}. \qquad [31]$$

These quantities measure provenance, calibration, and divergence. None is renamed as personal identity.

9.6 Typed domains, units, and partial results

The bookkeeping state in Equation [1] is heterogeneous. It must not be treated as one Euclidean vector whose raw coordinates can be added. Formally, the state belongs to a product of typed spaces:

$$X(t) \in \mathcal{X} = \mathcal{A}\times\mathcal{M}\times\mathcal{C}\times\mathcal{W}\times\mathcal{D}\times\mathcal{B}\times\mathcal{H}\times\mathcal{S}\times\mathcal{E}. \qquad [32]$$

The spaces have different objects, units, and natural timescales.

Family	Example mathematical object	Example raw units	Required comparison rule
A, anatomy	attributed graph plus spatial geometry	counts, micrometres, connection class	graph, geometry, and annotation distances remain separate
M, molecular state	positive concentration or count field	mol/L, molecules, normalized counts	assay-specific uncertainty and calibration
C, cellular physiology	state vector or response operator	volts, amperes, siemens, seconds	compare matched protocols, not isolated labels
W, synaptic/plasticity state	signed weighted edges plus update state	siemens, release probability, dimensionless optimizer state	sign, magnitude, delay, and rule must be declared
D, neural dynamics	multivariate time series or state trajectory	volts, spikes/s, phase, seconds	sampling rate, alignment, and horizon must be frozen
B, body state	kinematic, interoceptive, and actuator state	metres, radians, newtons, physiological units	body-specific coordinate map and latency
H, history	ordered provenance graph	event identifiers and timestamps	event ancestry, order, and uncertainty
S, self-model	calibrated predictions and categorical claims	probabilities, proper-score units, labels	score accuracy, calibration, and provenance separately
E, environment	world state and relational context	task-specific physical units and identifiers	held-out world and relationship split

Every family therefore needs a preregistered dimensionless scaling map before distances are combined. A scale may be a repeated-measurement floor, a source range, or another physically justified normalization; it cannot be chosen after seeing which successor wins. Weights are nonnegative and dimensionless. A weighted score is undefined when its components have not been brought into compatible direction and scale.

The evidence ladder is also a *partial* vector. Its phenomenal and numerical-identity entries are not forced into $[0,1]$ merely to make a chart complete:

$$\mathbf{L} \in ([0,1] \cup \{\perp\})^8 , \qquad \perp = \text{unassigned under the declared evidence and criterion.} \qquad [33]$$

The same rule applies to any technical rung whose anchor collapses or whose data are missing. `Unassigned` is not zero, failure, or success.

Equation [21] is a scalar only if every B_k has first been converted to one common resource unit. Storage bits, acquisition seconds, energy, financial cost, tissue damage, and ethical burden cannot be directly added. The general capture budget is therefore a vector, with an optional scalarization fixed before evaluation:

$$\begin{aligned} &\mathbf{B}_{cap} = (b_{storage}, b_{time}, b_{energy}, b_{money}, b_{risk}), \\ &B^{(c)}_{cap} = \sum_r \lambda^{(c)}_r b_r, \qquad [\lambda^{(c)}_r] = \frac{\text{common cost unit}}{[b_r]}. \end{aligned} \qquad (34)$$

Different scalarizations c represent different resource policies and must not be switched after results are known.

9.7 Endpoint-specific state sufficiency

The central scientific question is not whether every microscopic detail was copied. It is which captured variables are sufficient for a named endpoint, under a named intervention and time horizon. Let K index a capture package, Y_K its measured data, U_j the inputs and interventions for endpoint j, and O_j the held-out outcome. For a preregistered proper loss ℓ_j, define the best achievable predictive risk from that package as

$$R_j(K) = \inf_{f \in \mathcal{F}_K} \mathbb{E}[\ell_j(O_j, f(Y_K, U_j))] . \qquad [35]$$

The model class $\mathcal{F}_K$, training budget, and information available to it are frozen. Package K is *operationally sufficient* for endpoint j only relative to a full-state oracle or strongest feasible reference package K_{full}, an equivalence margin ϵ_j, and uncertainty level α:

$$K \succeq_{j,\epsilon_j,\alpha} K_{full} \iff \mathrm{UCB}_{1-\alpha}(R_j(K) - R_j(K_{full})) \leq \epsilon_j. \qquad [36]$$

This is a finite-domain empirical claim, not proof that omitted variables are universally irrelevant. A capture may be sufficient for route choice over ten seconds and insufficient for autobiographical calibration after a body transfer. Several different packages may also be sufficient because their information is redundant.

When a joint distribution is known, the residual information in omitted state $X_{\bar{K}}$ about endpoint O_j can be described by

$$I(O_j; X_{\bar{K}} \mid Y_K, U_j) \leq \eta_j. \qquad [37]$$

Conditional mutual information is a diagnostic, not an automatic biological sufficiency proof. It depends on the sampled intervention distribution, model support, and measurement quality. In biological use, the full latent state is unavailable, so Equation [37] can normally be estimated only in synthetic systems or bounded under explicit assumptions. Shannon's information measure supplies the mathematical ancestry, not an identity criterion [51].

The corresponding necessity question is incremental and conditional. If $V_j(K) = -R_j(K)$, the value of adding family k to package K is

$$\Delta V_{j,k|K} = V_j(K \cup \{k\}) - V_j(K). \qquad [38]$$

Main effects are not enough when state families interact. The benchmark must include specified pairs and higher-order combinations where mechanism predicts synergy. A null $\Delta V_{j,k|K}$ under adequate power is retained as a local negative result. It says that family k did not earn extra complexity for endpoint j, package K, and tested regime; it does not convert a local ablation into a universal state claim.

9.8 Structural and practical identifiability

Let θ parameterize a source or reconstruction model, and let $\mathcal{Q}$ be the allowed safe query set. Full parameter recovery is not required when only an endpoint functional $h_j(\theta)$ matters. That functional is structurally identifiable under $\mathcal{Q}$ when

$$[P_\theta(Y \mid q) = P_{\theta'}(Y \mid q) \text{ for every } q \in \mathcal{Q}] \Longrightarrow h_j(\theta) = h_j(\theta'). \qquad [39]$$

If two source states generate the same permitted observations but different endpoint values, no estimator using only those observations can guarantee endpoint recovery. The honest result is an equivalence class of compatible states, not a falsely precise reconstruction.

For a locally linearized discrete system

$$x_{t+1} = Ax_t + Bu_t, \qquad y_t = Cx_t,$$

the finite-horizon observability matrix is

$$\mathcal{O}_H = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{H-1} \end{bmatrix}, \qquad \operatorname{rank}(\mathcal{O}_H) = n \text{ is required to identify an } n\text{-dimensional local state.} \qquad [40]$$

This rank test is a transparent local diagnostic from the observability tradition [45], not a claim that a brain is linear. In a two-state example with $C = [1\ 1]$ and $A = I$, every observation reports only $x_1 + x_2$, so $(1, 0)$ and $(0, 1)$ remain indistinguishable. With $A = \operatorname{diag}(1, 0.5)$, observations at two times produce rows $[1, 1]$ and $[1, 0.5]$; the determinant is -0.5, and the two coordinates become locally identifiable in the noiseless model. Time structure can therefore add information that a snapshot lacks.

Practical identifiability additionally asks whether finite noisy data distinguish the compatible states. For differentiable probabilistic models, use scaled parameters $\vartheta = S_\theta^{-1}(\theta - \theta_0)$ and the query-conditioned Fisher information

$$\mathcal{I}_\mathcal{Q}(\vartheta) = \mathbb{E}\Big[\nabla_\vartheta \log p(Y \mid \vartheta, \mathcal{Q}) \nabla_\vartheta \log p(Y \mid \vartheta, \mathcal{Q})^T\Big]. \qquad [41]$$

The smallest singular value, condition number, profile likelihood, and posterior coverage must be reported. A singular or nearly singular matrix means some directions are unresolved at the tested scale; under regularity conditions this is the local-identification role of the information matrix formalized by Rothenberg [46]. A large matrix rank produced only by arbitrary parameter units is invalid; scaling must be frozen and physically justified. For non-differentiable models, simulation-based calibration and held-out intervention discrimination replace Fisher-information claims.

Adaptive querying earns a sufficiency claim only if posterior calibration is tested on known synthetic systems, safe-query support covers the relevant alternatives, and a deliberately misspecified candidate ensemble is included. Entropy can decrease around the wrong model when the true model is absent.

9.9 Error propagation, stability, and timescales

Raw errors from different state families are first mapped into a dimensionless norm $\|z\|_S = \|S^{-1}z\|$ using a frozen scale operator S. Let $\delta_t = \|X_t - \widetilde{X}_t\|_S$. If the source transition is locally Lipschitz with constant L_t, and ρ_t bounds reconstruction-model, input, body, environment, and numerical mismatch over one step, then

$$\delta_{t+1} \leq L_t \delta_t + \rho_t. \qquad [42]$$

Iteration yields the finite-horizon bound

$$\delta_T \leq \left(\prod_{i=0}^{T-1} L_i\right) \delta_0 + \sum_{j=0}^{T-1} \left(\prod_{i=j+1}^{T-1} L_i\right) \rho_j. \qquad [43]$$

For constant L and ρ, this reduces to

$$\delta_T \leq \begin{cases} L^T \delta_0 + \rho(1 - L^T)/(1 - L), & 0 \leq L < 1, \\ \delta_0 + T\rho, & L = 1, \\ L^T \delta_0 + \rho(L^T - 1)/(L - 1), & L > 1. \end{cases} \qquad [44]$$

Thus a small capture error need not remain small. In a contractive example with $\delta_0 = 0.02$, $L = 0.8$, $\rho = 0.005$, and $T = 10$, Equation [44] gives $\delta_{10} \leq 0.02446$. Changing only L to 1.05 gives $\delta_{10} \leq 0.09547$. The same starting accuracy supports different horizon claims in different stability regimes.

If an endpoint map g_j is locally Lipschitz with constant G_j, endpoint error obeys

$$d_j\Big(g_j(X_T), g_j(\widetilde{X}_T)\Big) \leq G_j \delta_T + \xi_{j,T}, \qquad [45]$$

where $\xi_{j,T}$ covers endpoint measurement and evaluator error. Conversely, a large microscopic trajectory distance can coexist with a small task error when many microstates implement the same functional invariant. State, trajectory, and endpoint errors must therefore all remain visible.

For potentially chaotic dynamics, report the finite-time divergence rate

$$\widehat{\lambda}_T = \frac{1}{T} \log\left(\frac{\delta_T + \epsilon_{num}}{\delta_0 + \epsilon_{num}}\right), \qquad [46]$$

with the perturbation scale, numerical floor ϵ_{num}, norm, and horizon. A positive finite-time rate limits exact trajectory replay; it does not by itself rule out distributional, attractor-level, task-level, or causal-response equivalence.

Each state family also has a characteristic timescale τ_k. With acquisition interval Δt_k and total capture window $T_{cap,k}$, register

$$r_k = \frac{\Delta t_k}{\tau_k}, \qquad w_k = \frac{T_{cap,k}}{\tau_k}. \qquad [47]$$

Values $r_k \ll 1$ indicate dense sampling relative to the process; $r_k \gtrsim 1$ warn that relevant changes may be averaged or aliased. Values $w_k \ll 1$ warn that the window is too short to characterize variation on that timescale. The thresholds are process- and estimator-specific. A Nyquist statement is allowed only for a justified approximately band-limited signal. Anatomy, receptor trafficking, membrane voltage, oscillatory phase, body state, and autobiographical learning cannot share one default sampling interval.

Time warping in Equation [29] is also bounded by mechanism. The allowed γ must be monotone, preserve registered event order, respect maximum latency and dilation, and expose both raw-time and warped-time results. Otherwise a reconstruction could appear accurate by stretching away missing responses.

9.10 Estimators, uncertainty, anchor sensitivity, and power

The clipped normalizations in Equations [25] and [26] are useful for displays but can hide magnitude beyond the anchors. For a lower-is-better metric, retain the raw value, the *unclipped* normalized value, and the clipped display value:

$$\bar{e} = \frac{\ell - \ell^*}{\ell^0 - \ell^*}, \qquad e = \mathrm{clip}_{[0,1]}(\bar{e}), \qquad \frac{\partial \bar{e}}{\partial \ell} = \frac{1}{\ell^0 - \ell^*}. \qquad [48]$$

Inside the unclipped region,

$$\frac{\partial \bar{e}}{\partial \ell^*} = \frac{\ell - \ell^0}{(\ell^0 - \ell^*)^2}, \qquad \frac{\partial \bar{e}}{\partial \ell^0} = -\frac{\ell - \ell^*}{(\ell^0 - \ell^*)^2}. \qquad [49]$$

These derivatives diverge when the anchors approach one another. Define an anchor-separation diagnostic

$$A_m = \frac{|\widehat{a}_m^0 - \widehat{a}_m^*|}{\sqrt{\widehat{\mathrm{Var}}(a_m^0) + \widehat{\mathrm{Var}}(a_m^*) - 2\widehat{\mathrm{Cov}}(a_m^0, a_m^*)}}. \qquad [50]$$

If the denominator is zero or A_m falls below a preregistered stability threshold, the normalized metric is unresolved. There is no universal threshold; it must be chosen from pilot calibration. For example, anchors 0.10 and 0.50 give sensitivity 2.5 per raw-loss unit, whereas anchors 0.10 and 0.12 give sensitivity 50. The latter score can reverse under small anchor error and cannot be treated as stable merely because clipping returns a number.

For one fixed metric and capture condition, let d_{srbwe} denote a per-episode result from source agent s, reconstruction seed r, body condition b, held-out world w, and episode e. The estimand is defined by nested averaging over the frozen evaluation distribution, while uncertainty retains the hierarchy:

$$\begin{aligned} d_{srbwe} &= \mu + u_s + v_{r(s)} + z_b + h_w + \varepsilon_{srbwe}, \\ \mathrm{Var}(d) &= \sigma_s^2 + \sigma_r^2 + \sigma_b^2 + \sigma_w^2 + \sigma_e^2 \\ &\quad + \text{registered interaction terms.} \end{aligned} \qquad (51)$$

This random-effects expression is a planning model, not permission to assume Gaussian independent residuals. The primary interval remains an outer source-cluster bootstrap with reconstruction seeds, worlds, and episodes resampled inside source. Report all variance components or explain why one is not estimable. Evaluator variance is added when human scoring is used. Cluster-bootstrap behavior can be poor with few independent clusters, so the pilot must disclose the source count and compare interval methods rather than treating repeated episodes as independent evidence [49].

Repeated episodes do not substitute for independent source agents. If a source contributes m correlated observations with intracluster correlation ρ_{ICC}, the approximate design effect and effective observation count are

$$D_{eff} = 1 + (m-1)\rho_{ICC}, \qquad n_{eff} = \frac{n_{raw}}{D_{eff}}. \qquad [52]$$

Power is planned at the source-cluster level. For an approximate two-one-sided equivalence test on one standardized contrast with true mean μ, symmetric margin $\epsilon > |\mu|$, source-level standard deviation σ_s, familywise allocation α/M, and target power $1-\beta$, a conservative planning approximation is

$$N_s \gtrsim \left[\frac{(z_{1-\alpha/M} + z_{1-\beta})\sigma_s}{\epsilon - |\mu|}\right]^2. \qquad [53]$$

Pilot estimates determine σ_s and feasibility; the margin is set from scientific consequences, not chosen to fit the observed variance. If $\epsilon \leq |\mu|$, no sample size can make that assumed effect equivalent. The two-one-sided-test logic is inherited from equivalence testing rather than ordinary failure-to-reject reasoning [47]. Load-bearing metrics use simultaneous intervals through a preregistered Holm, max-T, or other justified familywise procedure [48]. Equivalence requires every critical endpoint to pass; a favorable average cannot compensate for a failed rung.

Missingness is classified as planned absence, technical failure, safety stop, attrition, or unresolved provenance. For a $[0,1]$ score with missing fraction p, a transparent assumption-free bound is

$$\mu \in \left[(1-p)\bar{y}_{obs},\ (1-p)\bar{y}_{obs} + p\right]. \qquad [54]$$

Model-based imputation may be reported only beside this bound and with a sensitivity analysis. Runs that fail, time out, or trigger a safety stop remain in the denominator under the preregistered rule.

The causal-continuity score also carries audit uncertainty. If V is verified edge weight, U unresolved edge weight, and W total required weight, then

$$C_{causal} \in \left[\frac{V}{W}, \frac{V+U}{W}\right]. \qquad [55]$$

The upper endpoint is not a favorable estimate; it is the most continuity compatible with unresolved evidence. Changing edge weights after observing a successor is prohibited.

Finally, information gain and query cost must have compatible dimensions. If entropy is measured in bits and raw cost $c(q)$ is measured in another unit, the adaptive utility is

$$U(q) = \widehat{\Delta H}_{bits}(q) - \lambda_c c(q), \qquad [\lambda_c] = \text{bits per cost unit}. \qquad [56]$$

Equation [56] is the dimensionally explicit form of Equations [20] and [30]. The conversion λ_c, safety set, stopping rule, and calibration criterion are frozen before querying.

9.11 Worked countermodels and adverse-result retention

The benchmark must try to produce the following failures, not merely successful-looking reconstructions.

1. **Snapshot nonidentifiability.** Two latent states with the same $x_1 + x_2$ pass a one-time readout but diverge after dynamics reveal their components. This tests whether added temporal queries repair the ambiguity.
2. **Graph match with causal mismatch.** Two one-edge networks have identical unweighted topology. One has gain $+1$, the other -1. Their edge F1 is one, but the response to the same positive intervention has opposite sign. The graph score passes while intervention-response equivalence fails.
3. **Stable task with drifting carrier.** Internal coordinates diverge while held-out task performance and causal response remain inside margins. This is evidence that the tested microscopic coordinates were not necessary for that endpoint, not evidence that no state matters.
4. **Accurate snapshot in an expansive regime.** A successor begins with small δ_0 but fails at the registered horizon because $L > 1$ or persistent model error accumulates. Short-horizon success is retained and not advertised as durable reconstruction.
5. **Rich model with no incremental value.** Under matched capacity and adequate power, $\Delta V_{j,k|K}$ remains inside a null-value region. The added family receives a local null disposition for endpoint j; the result is not removed from plots or tables.
6. **Rich model harmed by finite data.** The fuller model overfits and performs worse on held-out sources or worlds. Capacity matching, regularization, and sample-efficiency curves distinguish model flexibility from captured-state value.
7. **Confident but misspecified query policy.** Posterior entropy decreases while oracle coverage and calibration fail because the true generator is outside the candidate ensemble. Confidence is scored as error, not as recovered state.
8. **Warping-dependent success.** Warped trajectory distance passes while raw latency, event order, or maximum dilation fails. Both values remain reported, and the dynamical rung stays open under its frozen rule.
9. **Right-censored branching.** Two successors do not cross the divergence threshold during observation. The result is right-censored at the horizon; it is not recorded as infinite sameness or numerical identity.
10. **Missing-not-at-random success.** Hard cases disproportionately time out. The complete-case mean may look favorable, but Equation [54] and the failure ledger expose the possible range.

These adverse and null outcomes are part of the scientific product. They remain in the run bundle, tables, uncertainty calculation, and narrative interpretation.

9.12 Source-faithful anti-strawman gate

The 2012-2024 SAN sources do not say that one fixed list of microscopic variables must be duplicated exactly before any function can survive. They describe an embodied, changing, memory-bearing network whose current operation depends on receiver state, recurrent activity, body-world return, and history; they also propose reading a running system through controlled interaction. Mathematical-Specification Stage translates that argument into candidate state families, endpoint-specific ablations, active queries, and trajectory tests.

The external literature is represented at equal resolution. Connectomes can carry substantial predictive structure. Physiology, perturbation, molecular identity, body dynamics, and learned state can add constraints. Neither observation is compressed into the slogan that topology is sufficient for every endpoint or useless for every endpoint. The benchmark asks where each variable earns incremental predictive or causal value.

Accordingly, the following reviewer moves are inadmissible:

- turning Equation [1] from a search space into a universal exact-copy requirement;
- treating the 2017 hard-drive analogy as a static-disk claim while omitting its current-state and difference-only operations;
- treating immediate copy divergence as denial of inherited similarity;
- treating a local null ablation as universal irrelevance;
- treating nonidentifiability under one query set as impossibility under every query set;
- treating behavioral success, calibrated self-report, or causal continuity as a theory-neutral identity or consciousness verdict; or
- removing adverse results because they do not support a binary upload narrative.

An apparent mismatch first triggers reconstruction of the complete source claim, its scale, timescale, receiver, endpoint, assumptions, and scientific comparator. Open identifiability, sufficiency, phenomenal, and identity questions remain open work; they are not rewritten as claims the source did not make.

9.13 Implementation Freeze

The Deterministic Reference Implementation may implement the source agent, reconstructor, schema validator, fixtures, and deterministic tests only after it binds the following Mathematical-Specification Stage additions in a versioned schema amendment:

- typed units and scaling maps for every state family and metric;
- raw, unclipped, and clipped normalized scores;
- anchor-separation status and collapsed-anchor handling;
- endpoint, horizon, intervention set, equivalence margin, and multiplicity family;
- omitted-state partition and conditional ablation design;
- identifiability diagnostic and unresolved equivalence class;
- raw-time and bounded-warp trajectory results;
- stability and timescale metadata;
- source-cluster, reconstruction, body, world, episode, and evaluator identifiers;

- complete failure, timeout, safety-stop, and missingness classes;
- right-censoring for branch divergence; and
- unresolved-edge intervals for causal continuity.

No controlled result is claimed at Mathematical-Specification Stage. The application remains specified, not implemented or run. The new mathematics constrains what Deterministic Reference Implementation must validate and what later runs are allowed to mean.

9.14 Executable Reference Implementation

The Deterministic Reference Implementation converts the frozen specification into a bounded application without treating an implementation check as evidence that a mind has been uploaded. The ordinary engineering problem is straightforward: a benchmark cannot be audited if its source state, reconstruction condition, metrics, failure states, and provenance are only prose. The solution is to create an artificial source agent whose complete state is known, reconstruct it under the declared C0-C10 ladder, emit typed result bundles, and try to break those bundles with validators and counterexamples. Only after that operation is concrete is the term **Upload Ladder Benchmark reference implementation** used [52].

9.14.1 Source agent and reconstruction pipeline

The deterministic source agent carries all nine candidate state families defined in Equation [1]: attributed graph and geometry (A); receptor-like gain state (M); membrane-like bias state (C); signed recurrent weights (W); ongoing hidden state (D); body transfer function (B); event history (H); calibrated self-model variables (S); and world-model parameters (E). These variables are dimensionless reference quantities except where the schema names graph structure, micrometre geometry, event identifiers, steps, or proper-score units. They are not measurements of a biological person.

For source seed (s), reconstruction seed (r), capture condition (k), body condition (b), world seed (w), and horizon (T), the executable pipeline is

1. generate and hash the complete artificial source state;
2. construct the exact included and omitted state-family partition for $k \in \{C0, \ldots, C10\}$;
3. reconstruct a successor from demonstrations, a snapshot, a staged transfer, or a branching condition according to the frozen map;
4. optionally change the successor body transform while retaining the body-condition label;
5. run source and successor through the same seeded world sequence;
6. apply the registered hidden-state, weight, memory, and body interventions;
7. compute MU-M01 through MU-M15, preserving raw values, units, normalization anchors, unclipped and clipped values, horizons, margins, missingness, and identifiability status;
8. emit ordered provenance and failure events; and
9. hash the canonical result bundle.

The implementation is deterministic: repeating the same input tuple produces the same state hashes, event hashes, and result-bundle hash. This property supports auditing. It does not imply that biological development, neural recording, or personal identity is deterministic in the same way.

9.14.2 Additive typed schema and validation

Schema version 0.3.0 is an additive successor to the Source-and-Citation Stage contract. It binds every Mathematical-Specification Stage freeze item: units and positive scaling maps for (A,M,C,W,D,B,H,S,E); raw and normalized metric values; anchor-separation status; endpoint, horizon, safe intervention set, alpha, margin, and multiplicity family; the included/omitted partition; identifiability rank and compatible-state class; finite-horizon stability and timescale ratios; source, reconstruction, body, world, episode, and evaluator identifiers; failure and missingness classes; branch right-censoring; and the causal-continuity interval.

Validation occurs at two levels. A JSON Schema checks required fields, types, enumerations, array sizes, and hashes when the optional standards library is available. A deterministic application validator independently enforces cross-field invariants. It rejects overlapping or incomplete state partitions, a partition inconsistent with its C-condition, duplicate or missing metrics, an observed metric without a raw value, a normalized value inconsistent with its anchors, a collapsed anchor that still emits a score, impossible identifiability ranks, incorrect stability bounds, discontinuous event sequences, missing retained-failure events, and mismatched provenance or bundle hashes. The application validator is not a weaker fallback: it protects benchmark-specific relations that a field-level schema alone cannot express.

9.14.3 Development fixtures and adversarial fixtures

The development grid contains two artificial source agents, two reconstruction seeds, two held-out development worlds, two body conditions, and all 11 capture conditions. This produces 22 deterministic development bundles, 330 metric records, and 442 provenance events. Two MU-M10 observations are correctly retained as right-censored branch outcomes. All 22 bundles pass both the field and cross-field validators.

Twelve deliberately corrupted fixtures test the negative path. They remove a required source object, use an invalid checkpoint hash, overlap the state partition, leave the partition incomplete, mismatch condition and partition, remove a raw value from an observed metric, alter a normalized score, emit a score after anchor collapse, duplicate a metric identifier, make identifiability rank exceed dimension, corrupt the event-log hash, or corrupt the bundle hash. All 12 are rejected. This is evidence that the implementation detects these declared defects; it is not evidence that it detects every possible software or scientific error.

9.14.4 Countermodels and mathematical stress cases

All ten countermodels from Section 9.11 are executable and retain their adverse interpretation. Snapshot-equivalent latent states diverge under a temporal query. Identical unweighted graphs produce opposite intervention responses when edge sign changes. Task output remains stable while carrier coordinates drift. A small initial error crosses a registered margin in an expansive regime. A richer state family can add no endpoint value or harm held-out performance under finite data. A misspecified query policy becomes confident while losing oracle coverage. Warped trajectory distance can pass while raw latency fails. Branch divergence can remain right-censored. Missing-not-at-random failures can make a complete-case mean look favorable while the assumption-free interval remains wide. All ten checks pass because they successfully produce and retain the specified failure, not because every reconstruction succeeds.

The 12 Mathematical-Specification Stage stress cases also pass at their exact registered values. They cover lower-better normalization, collapsed anchors, contractive and expansive finite-horizon bounds, unobservable and observable two-state systems, graph/causal mismatch, missing-score bounds, causal-continuity intervals, cluster design effects, right-censoring, and a local null incremental-value result. Their meaning remains local. For example, the observable two-state case is a noiseless linear diagnostic, the propagation value is an upper bound rather than an observed error, and a local null does not establish universal irrelevance.

9.14.5 Provenance, failure retention, and claim boundary

Every bundle contains ordered events for source creation, capture, reconstruction, evaluation, each metric, any retained failure, and result finalization. Payload hashes, source and successor state-family hashes, event-log hashes, and result-bundle hashes make later alteration detectable. The failure ledger has explicit classes for planned absence, technical failure, safety stop, attrition, unresolved provenance, and right-censoring. The causal-continuity result is retained as the interval in Equation [55], never converted into an identity probability.

The final-test plan is separately hashed and reports `sealed_unopened`, `dataGenerated: false`, `dataOpened: false`, and `unsealCount: 0`. No final source seeds, reconstruction seeds, worlds, checkpoints, or observations exist in Deterministic Reference Implementation. The 22 bundles are development fixtures used to validate wiring, formulas, events, and rejection behavior. They are not biological reconstructions, controlled upload experiments, consciousness tests, identity adjudications, or evidence that human mind uploading is feasible. A confirmatory study must freeze sample size, power, compute budget, code version, evaluator assignments, stopping rules, and an unsealing procedure before creating final-test data.

9.15 Preregistered Synthetic Development Pilot

The next problem is not whether the reference implementation can execute one valid fixture. It is whether a balanced artificial population can expose stable, unstable, null, adverse, and censored outcomes without changing the rules after seeing them. Compact Development Pilot therefore freezes the source count, seeds, state access, capacity, interventions, evaluation horizon, primary contrast, margins, multiplicity, bootstrap, variance analysis, resource ceilings, stopping rules, and failure treatment before generating pilot data [53]. Only after those constraints are fixed is this stage called a **preregistered synthetic development pilot**.

9.15.1 Source-level power and frozen grid

The artificial source agent, successor architecture, metric definitions, validators, and countermodels are byte-identical copies of Deterministic Reference Implementation. The independent outer unit is the source agent, not the episode or bundle. The primary family contains MU-M03, MU-M05, MU-M06, MU-M07, MU-M08, and MU-M13. With familywise alpha 0.05, six one-sided Bonferroni components, equivalence margin $\Delta = 0.15$, planning source-level standard deviation 0.20, and target power 0.80, the normal approximation requires 19 sources. Twenty-percent inflation produces the frozen 24-source design. This calculation plans a development pilot; it is not a guarantee of final power under a new biological or artificial population.

Each of 24 source seeds is crossed with 11 C0-C10 capture conditions, two reconstruction seeds, two body conditions, and two held-out development-world seeds. The design therefore contains $24 \times 11 \times 2 \times 2 \times 2 = 2{,}112$ bundles and $2{,}112 \times 15 = 31{,}680$ metric observations. All seeds are listed in the preregistration. No cell is added after outcome inspection.

For source s, primary metric m, reconstruction seed r, body b, and world w, the paired source-level contrast is

$$\bar{d}_{s,m} = \frac{1}{|R||B||W|} \sum_{r \in R} \sum_{b \in B} \sum_{w \in W} \left[P_m(C6, s, r, b, w) - P_m(C7, s, r, b, w)\right], \qquad [57]$$

where P_m places each non-descriptive metric on its registered higher-is-better scale. The bootstrap resamples the 24 source-level contrasts, never individual metric rows as though they were independent. With $\alpha_m = 0.05/6$, local equivalence is assigned only when

$$-\Delta < Q_{\alpha_m}(\bar{d}_m^*) \leq Q_{1-\alpha_m}(\bar{d}_m^*) < \Delta. \qquad [58]$$

An interval wholly beyond one margin is not equivalent. Every other case is unresolved. This rule prevents failure to find a difference from being renamed equivalence.

9.15.2 Exact capacity and data-access matching

Every successor has two controller nodes, four recurrent-weight slots, nine state-family containers, 15 metric slots, no training updates, a 24-step horizon, four interventions, one evaluator implementation, and a budget of 480 controller steps per bundle. Across the pilot this produces 1,013,760 controller steps. Reconstruction seed, body, world, horizon, evaluator, intervention set, and compute budget are exactly paired within each C6-C7 source comparison.

Data access is declared exactly. C0 receives no captured state; C1 receives four demonstration events; C2-C6 receive their registered nested subsets; C7 receives all nine state families; and C8-C10 use the full families under staged or branching modes. The primary C6-C7 contrast therefore asks the local incremental-value question created by access to S and E, with the other declared factors matched. It does not ask whether (S) and (E) are universally necessary in biological minds.

9.15.3 Execution, retention, and primary results

The runner completes all 2,112 planned bundles and validates each before admitting it to the raw ledger. Every bundle contributes all 15 metrics. No outcome-based stop occurs. The metric ledger retains 6,496 adverse rows, 477 rows in the frozen null band, 22,595 favorable rows, 1,904 observed descriptive rows, and 208 right-censored branch rows. All ten countermodels execute and retain their adverse or null interpretation.

The primary paired results are:

Metric	C6 minus C7 estimate	Simultaneous bootstrap interval	Disposition
MU-M03 held-out task performance	-0.089410	[-0.116753, -0.062500]	local equivalence
MU-M05 intervention-response performance	-0.148914	[-0.174952, -0.123102]	unresolved
MU-M06 self-model performance	-0.137400	[-0.137400, -0.137400]	local equivalence
MU-M07 memory-provenance performance	0.000000	[0.000000, 0.000000]	local equivalence
MU-M08 changed-body transfer performance	0.044083	[0.032678, 0.055289]	local equivalence
MU-M13 critical-shift robustness	-0.236979	[-0.388911, -0.080729]	unresolved

"Local equivalence" does not mean equality. MU-M03 favors C7 on average, but the complete simultaneous interval remains inside the preregistered ± 0.15 margin. MU-M05 crosses the negative margin, while MU-M13 is too wide to satisfy either equivalence or non-equivalence. Both remain unresolved rather than being forced into a favorable conclusion. MU-M06 is source-invariant in this compact generator, which explains its zero-width source bootstrap interval; this is a property of the artificial implementation, not evidence of biological certainty.

9.15.4 Descriptive variance decomposition

Because the full design is balanced, Compact Development Pilot reports descriptive main-effect sums of squares. For factor f, level g, group size n_g, group mean $\bar{y}_g$, and grand mean $\bar{y}$,

$$SS_f = \sum_{g \in f} n_g (\bar{y}_g - \bar{y})^2, \qquad \eta_f^2 = SS_f / SS_{total}. \qquad [59]$$

Interactions and all variance not assigned to main effects remain in the residual. These fractions are not causal variance components.

Capture condition accounts descriptively for 0.473 of MU-M03 total variation, 0.774 of MU-M05, 1.000 of MU-M06 and MU-M07 in this generator, 0.141 of MU-M08, and 0.274 of MU-M13. Source-agent fractions are 0.079, 0.031, 0.000, 0.000, 0.032, and 0.133 respectively. Body condition accounts for 0.173 of MU-M08, while its fractions on the other five primary metrics are below 0.01. Residuals including interactions are 0.438 for MU-M03, 0.189 for MU-M05, 0 for MU-M06 and MU-M07, 0.650 for MU-M08, and 0.590 for MU-M13. The large residuals on transfer and robustness warn against compressing those endpoints into one-factor stories.

9.15.5 Resource and stopping audit

The preregistered ceilings are 600 wall seconds, 600 CPU seconds, 256 MiB traced peak memory, 64 MiB raw-bundle output, and 85 MiB total results. The deterministic reproduction completes in

231.49 wall seconds and 228.98 CPU seconds, reaches 18,287,064 traced peak bytes, writes 36,836,641 raw-bundle bytes, and produces 43,560,279 total result bytes. Thus the resource vector satisfies

$$(t_{wall}, t_{cpu}, m_{peak}, b_{raw}, b_{total}) \preceq (600, 600, 268{,}435{,}456, 67{,}108{,}864, 89{,}128{,}960). \qquad [60]$$

The raw bundle SHA-256 is `4DB0FDA2D296143CD227A5B5EAB81DFE2C5F2AE624AF8BEFB18248E01F5278F1`; the metric-ledger SHA-256 is `58DA0C4C54C720F2C1CB98CD16F39907698B5507615EEFF32662E093D13DE261`. An exact rerun reproduces both hashes. The resource receipt is not used to remove a slow or unfavorable cell.

9.15.6 Final-test separation and evidence boundary

The Compact Development Pilot reads only the Deterministic Reference Implementation seal metadata needed to verify separation. The seal hash remains `6CB7F80F513735467DD5723D9D88FF2665AA09B61A6BC7AA8619154BAF82A1E3`; status remains `sealed_unopened`; no final data are generated or opened; and the unseal count remains zero. The Compact Development Pilot package contains no final-test directory.

The pilot establishes that this artificial source/reconstruction system can execute a preregistered, capacity-audited, source-clustered comparison while retaining unfavorable results. It does not establish that the nine state families are sufficient for a living brain, that they can be measured at required resolution, that a successor is conscious, that a person continued, or that human mind uploading is feasible.

10. Failure conditions and discriminators

The incremental-value failure test is defined in Equation [24]. Other decisive outcomes include:

- a structurally close reconstruction that fails causal perturbation tests;
- a behaviorally convincing successor with poor memory provenance or self-model calibration;
- a functionally equivalent controller that fails after body transfer;
- a gradual replacement that preserves performance but produces discontinuous internal organization;
- two successors with equal source similarity but incompatible later identities;
- an apparent "upload" whose success is explained by imitation from external training data rather than captured source state; and
- a consciousness indicator that tracks task performance but not the claimed conscious distinction.

Each result says which rung passed and which remains open.

11. Ethics, welfare, and governance

Technical success and moral status should not wait for one another. A system with credible memory, preference, distress, self-model, and future-directed behavior may warrant precaution even when phenomenal continuity is unresolved. Conversely, declaring a system "the same person" creates ownership, debt, consent, and legal risks that no behavioral benchmark can settle alone.

Every experiment should preserve:

- source consent and revocation;
- a complete record of captured variables and omissions;
- provenance for every memory or model update;
- separation between reversible interface settings and persistent self-change;
- protection against unauthorized copying, editing, deletion, and forced labor;
- independent welfare assessment for every active successor;
- conflict rules for multiple successors claiming one legal identity; and
- the right to report uncertainty rather than perform an assigned identity.

Customization requires special care. Adding a tool is different from rewriting memory, value, agency, or attachment. The difference should be measurable in state, behavior, persistence, reversibility, and welfare.

12. Strong alternatives

A connectome-sufficiency program asks how far structure plus cell sign and simple dynamics can go. Its success is an empirical achievement and a strong baseline [7,12]. A functionalist upload program evaluates whether causal organization is preserved independent of substrate [29]. A biological-continuity account gives special weight to the living organism. A psychological-continuity account emphasizes memory, intention, character, and causal linkage [28]. Gradual-replacement arguments emphasize continuous operation, while Wiley and Koene argue that graduality has no automatic metaphysical privilege [30]. Extended-mind accounts show how external resources can enter cognition without implying that every coupled resource becomes the same person [27].

The Upload Evidence Ladder does not choose among these views by definition. It makes each view declare the measured facts and additional premises it uses. That is the paper's main methodological contribution.

12.1 Nonduplication boundaries

This paper has a distinct experimental object: a source-to-successor capture event evaluated across the eight-rung ladder. *The Field-Cell Self* asks what physical operation constitutes a continuing embodied self. *The Network That Learns It Is a Network* asks how a distributed system acquires explicit knowledge of its own network-made self-model. *Can a Neuron Store Infinite Information?* bounds recoverable information in continuous biological morphology. *Telescoping Network Reinstatement* tests population-pattern reconstruction under drift. *The External Cortex Loop* and the Coadaptive Substrate Principle ask when reciprocal interfaces become functionally incorporated. *Measuring Candidate Machine-Consciousness Mechanisms in SAN* supplies theory-dependent tests relevant to the phenomenal-evidence rung. *Artificial Neurology as Developmental Engineering* studies the longitudinal construction and welfare governance of history-bearing artificial individuals.

Those papers provide components, baselines, or downstream interpretations. None replaces the present paper's joined comparison of declared state capture, discontinuous copy, staged replacement, one-to-many branching, held-out functional and causal equivalence, source-memory provenance, phenomenal-evidence boundaries, and numerical-identity criteria. Conversely, the Upload Ladder does not absorb their full identity mechanism, self-knowledge model, information-capacity analysis, drift estimator, interface-incorporation test, consciousness battery, or developmental lifecycle.

13. Discussion

The likely path to useful reconstruction will not be one perfect scan. It will combine anatomy, cell-type inference, physiology, perturbation, behavioral modeling, recurrent state estimation, embodiment, and longitudinal coadaptation. Some functions may tolerate aggressive abstraction. Others may depend on fine receiver state, body chemistry, or active history. The correct resolution is an empirical variable, not a slogan.

The 2013 SAN copy-divergence argument anticipated the need to distinguish inherited organization from situated continuation. Contemporary connectomics sharpens rather than erases that point. A wiring diagram can explain meaningful computations, yet a working organism also receives current evidence, acts, changes its body and environment, and learns. A reconstructed successor should therefore be evaluated as a new causal process with a declared relation to its source.

The phenomenal question remains harder. A system can pass increasingly demanding structural, dynamical, functional, causal, and self-model tests. Those results can increase the case for treating it as a conscious subject under stated theories. They do not provide a theory-neutral measurement of first-person continuity. The responsible result may be strong evidence for a new conscious successor and no settled evidence that the original person's experience continued.

14. Conclusion

Mind uploading becomes scientifically tractable when it stops being one binary promise. This paper offers a typed ladder from measurement to structure, dynamics, function, self-model, causal continuity, phenomenal evidence, and identity. It supplies equations for capture error, trajectory comparison, causal equivalence, branching, and adaptive neural query; recovers a 2011-2024 SAN genealogy without backdating later terms; and specifies an executable embodied benchmark.

The central rule is simple: report exactly what was captured, reconstructed, tested, and continued. A copied graph is a copied graph. A working model is a working model. A behaviorally and causally similar successor is a major result. Whether that successor is conscious, whether the source continued, and whether the law should treat them as one person remain additional questions with additional burdens of proof.

15. Architecture-Robustness Study

Study rationale

The Compact Development Pilot answered a narrow engineering question: can a frozen, typed upload-evidence benchmark be executed without dropping adverse, null, or censored outcomes? It could. That result did not show that its favorable C6-C7 findings were robust to the artificial dynamics chosen for the source agent. A single compact recurrent system can accidentally make one declared state family look sufficient because of its own parameterization. The next problem is therefore ordinary and concrete: **does the conclusion survive when the same capture conditions are tested in a materially different recurrent architecture under the same capacity and evaluation budgets?**

Study A answers that question before introducing new metaphysical language. It implements two complete versions of their *declared artificial substitute models*: a gated recurrent substitute and an independently generated echo-state substitute. "Complete" here means that each stated 16-node model, intervention battery, metric pipeline, trace translator, and validator is implemented. It does **not** mean that either model is a complete brain, a faithful biological original, a conscious subject, or an upload system. Those boundaries are part of the executable contract, not an apologetic afterthought.

The vocabulary is earned in this order. First, the two systems must receive comparable inputs and expose comparable capacities. Second, the capture conditions must control which state families are copied. Third, the systems must be tested under ordinary and perturbed trajectories. Fourth, the model-specific estimates must be compared. Only then can the study ask whether an observed C6-C7 relation is **architecture-robust**.

15.1 Two bounded recurrent substitutes

Both substitutes contain 16 hidden nodes, 256 recurrent-weight slots, 64 input-weight slots, 16 readout-weight slots, nine declared state families, no training updates, and the same 32-step evaluation horizon. Their recurrent operators differ.

For the gated recurrent substitute,

$$\tilde{h}_{t+1} = \tanh(Wh_t + Ux_t + vg_t + c + r_t + 0.05s)\,, \qquad h_{t+1} = (1-m)\odot h_t + m \odot \tilde{h}_{t+1}. \tag{61}$$

Here $h_t \in \mathbb{R}^{16}$ is the hidden state; $W \in \mathbb{R}^{16\times16}$ is the recurrent matrix; $U \in \mathbb{R}^{16\times4}$ maps the four-dimensional observation; $v \in \mathbb{R}^{16}$ maps the scalar goal g_t; $c, r_t \in \mathbb{R}^{16}$ are bias and object-memory terms; s is a dimensionless self-state summary; and $m \in [0,1]^{16}$ is a source-specific update gate. All model quantities are dimensionless synthetic variables.

For the echo-state substitute,

$$q_{t+1} = \tanh(Wh_t + Ux_t + vg_t + c + r_t + 0.05s)\,, \qquad h_{t+1} = (1-m)\odot h_t + m\odot \tanh(q_{t+1} + 0.15h_t). \tag{62}$$

The same symbol m now acts as a leak-rate vector. The recurrent matrices are generated from different architecture-specific seeds, normalized to bounded row sums, and retained in source and successor checkpoint hashes. The models share dimensionality and budget, but not recurrent geometry or update law. This is a capacity-matched two-architecture test, not a claim that the systems reproduce any named biological circuit.

15.2 Frozen grid, access contract, and provenance

The preregistration was frozen before result generation at SHA-256 `4B1FC597B79DB0152E8A5D081F6F24F44A70392648691FD9EA0EBFA5B5BA5547`. The grid contains 16 source seeds, two architectures, five capture conditions (C1, C3, C5, C6, and C7), two reconstruction seeds, two body conditions, and two world seeds:

$$N_{\text{bundle}} = 16 \times 2 \times 5 \times 2 \times 2 \times 2 = 1{,}280. \tag{63}$$

Every bundle emits 15 metrics, two checkpoint rows, 64 translated baseline-trace rows, and 12 paired intervention rows. The translated trace stores action, mean, variance, norm, positive fraction, maximum absolute activation, sign pattern, and the SHA-256 of the underlying 16-value state. It deliberately does not store or claim to decode phenomenology.

The capture-family contract is unchanged in meaning: C1 receives a bounded demonstration summary; C3 receives anatomy, cellular/bias state, recurrent weights, and current dynamics; C5 additionally receives modulation and history; C6 additionally receives embodiment; and C7 additionally receives self-model and environment/interface state. The intended contrast remains C6 minus C7. The difference is that Study A tests it separately in both artificial architectures.

15.3 Expanded adversarial battery and controller budget

Each source-successor pair is challenged by 12 prespecified interventions: hidden impulse, recurrent-weight silence, body remapping, memory occlusion, recurrent-sign inversion, quarter-node dropout, observation noise, goal reversal, one-step latency, self-model corruption, world/input-gain shift, and alternating bias shift. For intervention j, the discrepancy is

$$d_j = \frac{1}{2} d_{\text{trajectory},j} + \frac{1}{2} L_{\text{action},j}, \qquad 0 \le d_j \le 1, \tag{64}$$

where the trajectory term is the normalized root-mean-square hidden-state difference and the action term is the mismatch fraction. Passing at the frozen local margin requires $d_j \le 0.15$. A pass is an agreement result inside the artificial model, not a safety or consciousness result.

The exact controller-step budget per bundle is reconciled rather than estimated:

$$B_H = \big(2 + 2J + 1 + 1 + 2 + 2 + 1\big)H = 33 \times 32 = 1{,}056, \tag{65}$$

where $J = 12$ paired interventions; the remaining terms are the baseline source-successor pair, same-body comparison, damage comparison, deterministic replay pair, reverse-sequence replay pair, and one unscored diagnostic successor trace. Across all bundles this is 1,351,680 controller steps.

15.4 Architecture-specific equivalence and difference-in-differences

For source cluster a, architecture k, and metric m, the source-level contrast is

$$\Delta_{akm} = \overline{Y}_{akm,C6} - \overline{Y}_{akm,C7}. \tag{66}$$

The analysis resamples the 16 source clusters, not the 128 repeated condition cells, for 3,000 fixed-seed bootstrap draws. With four primary metrics, the one-sided alpha per metric is $0.05/4 = 0.0125$. Local equivalence requires the complete familywise interval to lie inside $(-0.15, 0.15)$. If the interval crosses a margin, the result is unresolved. If it lies beyond a margin, the result is non-equivalent.

Architecture sensitivity is measured by

$$\Gamma_{am} = \Delta_{a,\text{gated},m} - \Delta_{a,\text{echo},m}. \tag{67}$$

Study A reports Γ and the two architecture-specific dispositions. It does not pool them when their dispositions disagree. That rule prevents a favorable architecture from averaging away a failure in the other.

15.5 Results: The Compact-Pilot Equivalence Result Does Not Replicate

All 1,280 planned bundles completed. The validators accepted 1,280/1,280 bundles. The package retains 19,200/19,200 metric rows, 15,360/15,360 paired intervention rows, 2,560/2,560 checkpoint rows, and 81,920/81,920 translated trace rows.

Architecture	Metric	C6-C7 estimate	Familywise bootstrap interval	Disposition
gated recurrent substitute	MU-M03 held-out task performance	-0.240234	[-0.285889, -0.190060]	not equivalent
gated recurrent substitute	MU-M05 intervention performance	-0.283700	[-0.304430, -0.261312]	not equivalent
gated recurrent substitute	MU-M08 changed-body transfer	0.135383	[0.079411, 0.193226]	unresolved
gated recurrent substitute	MU-M13 adversarial pass fraction	-0.651042	[-0.746094, -0.571615]	not equivalent
echo-state substitute	MU-M03 held-out task performance	-0.181885	[-0.239627, -0.122427]	unresolved
echo-state substitute	MU-M05 intervention performance	-0.240347	[-0.266100, -0.215566]	not equivalent
echo-state substitute	MU-M08 changed-body transfer	0.167300	[0.129783, 0.207060]	unresolved
echo-state substitute	MU-M13 adversarial pass fraction	-0.606120	[-0.671224, -0.544271]	not equivalent

No architecture-specific primary endpoint meets the frozen local-equivalence rule. Five of eight contrasts are non-equivalent and three are unresolved. MU-M03 produces the required cross-model disagreement: the gated model is non-equivalent while the echo-state model is unresolved.

Consequently, Study A issues no pooled MU-M03 label. MU-M05 and MU-M13 are non-equivalent in both architectures; MU-M08 is unresolved in both.

The difference-in-differences intervals for MU-M03, MU-M05, and MU-M08 lie within the local architecture margin. Their respective estimates are -0.058350, -0.043352, and -0.031917. MU-M13 remains architecture-sensitivity unresolved because its interval [-0.167318, 0.070313] crosses the negative margin. These comparisons show that both models expose the same broad C6-C7 failure direction while still differing enough to forbid a universal architecture claim.

This is the opposite of a promotional benchmark result. It is useful because it identifies which conclusion from the Compact Development Pilot was model-dependent. Under these stronger substitutes, adding the self-model and environment/interface families in C7 materially changes the measured endpoints. The result supports retaining those families as live sufficiency candidates in this artificial test. It does not prove that the same variables are necessary or sufficient in a biological brain.

15.6 Adverse, null, censored, and intervention evidence

The complete 19,200-row metric ledger contains 6,007 adverse rows, 511 null-band rows, 134 right-censored rows, 1,146 descriptive observed rows, and 11,402 favorable rows. Exactly 6,652 adverse, null, or censored rows are retained, with zero unfavorable omissions. The intervention ledger contains 2,026 passes and 13,334 failures at the 0.15 discrepancy margin. Those failures are part of the result, not implementation debris.

The right-censored rows are time-to-divergence observations in which no action divergence occurs within 32 steps. They are not converted into observed divergence times. The 32-step horizon is an explicit unit of the estimate; extending the horizon could change the censoring pattern.

Ten operational comparison surfaces are reported separately for each architecture: demonstration-only imitation, structural snapshot, memory-augmented snapshot, self/environment-omitted snapshot, full-state snapshot, recurrent-sign inversion, quarter-node dropout, self-model corruption, observation latency, and world/input-gain shift. These are baselines and ablations within the two artificial substitutes. They are not faithful implementations of biological, philosophical, or competitor theories.

15.7 Stability, timescale, and internal-state provenance

Every baseline hidden state is finite across the bounded horizon. The largest observed absolute hidden activation is 0.899643. The largest source and successor recurrent spectral radius is 0.730803. These values are consistent with bounded behavior over the tested 32 steps, but they do not prove asymptotic stability, robustness under arbitrary inputs, or biological plausibility.

For each source and successor, the checkpoint ledger stores nine family hashes and a whole-checkpoint hash. The trace ledger stores 81,920 translated state records with raw-state hashes but not raw internal vectors. Thus an auditor can verify that a trace belongs to a particular simulated state without treating the translated features as a semantic or phenomenal readout. The ordered result-bundle hashes are themselves hashed into one completion receipt.

15.8 Descriptive variance and uncertainty boundaries

For each primary metric, Study A partitions balanced descriptive sums of squares across source, architecture, capture condition, reconstruction seed, body, world, three prespecified two-way interactions, and a residual containing higher-order and unassigned variation:

$$SS_{\text{total},m} = \sum_r (Y_{rm} - \overline{Y}_m)^2, \qquad \rho_{qm} = \frac{SS_{qm}}{SS_{\text{total},m}}. \tag{68}$$

Capture condition explains the largest declared fraction for MU-M05 (0.5944) and MU-M13 (0.6387). For MU-M03 and MU-M08, the residual containing higher-order and unassigned variation is largest (0.6058 and 0.6970). The capture-by-body interaction is also material for MU-M03 (0.1673), MU-M08 (0.1599), and MU-M13 (0.2056). These are descriptive balanced sums of squares, not causal variance components or random-effects estimates. The result warns against treating capture condition as the only source of uncertainty.

15.9 Resource reconciliation and deterministic validation

The run consumed 340.58 wall seconds and 339.03 CPU seconds, with a traced peak of 16,824,763 bytes. Raw bundles occupy 9,657,233 bytes; translated traces occupy 30,166,920 bytes; all result files, including the settled resource receipt, occupy 51,241,804 bytes. Every frozen resource cap passes.

The independent validator performs 37 checks: exact row counts; receipt hashes; bundle schemas; checkpoint family coverage and self-consistent hashes; deterministic sentinel regeneration; primary-analysis reproduction; architecture-specific reporting; final-seal preservation; adverse-result retention; stability fields; and resource limits. All 37 pass. The Study A test suite adds 28/28 passing tests. the internal manuscript lineage retain their prior SHA-256 values.

15.10 What Study A establishes—and what it does not

Study A establishes that:

- the frozen upload-evidence pipeline can execute two capacity-matched artificial recurrent architectures;
- checkpoint and translated internal-state provenance can be audited at scale;
- all 12 interventions and all unfavorable outcomes can be retained without selective stopping;
- architecture-specific cluster bootstrap and difference-in-differences can be reproduced; and
- The favorable Compact Development Pilot C6-C7 equivalence conclusion does not survive the stronger Study A models.

Study A does not establish that either substitute is a faithful brain model; that the nine state families are complete; that a human brain can be measured at the required resolution; that a reconstructed system would be conscious; that a numerical person continues; or that mind uploading is feasible. It also does not establish long-horizon stability, generalization outside the deterministic world generator, or robustness to learned model misspecification.

The next honest gate is not publication of an upload-success claim. It is an independently implemented, preregistered replication using a higher-capacity learned recurrent system and a biologically constrained network, with checkpointed training provenance, held-out environments, blinded evaluation, and the same no-omission rule. The existing final-test plan remains byte-identical and `sealed_unopened`; Study A performs zero final-data reads, zero final-data writes, and zero unseals.

16. Learned-Model Replication

16.1 The adverse result remains the starting point

Study B begins with, rather than erases, the strongest prior result. Study A found **zero** model-specific local-equivalence endpoints, **five** non-equivalent endpoints, **three** unresolved endpoints, and one cross-model disagreement for which pooling was prohibited. It retained 6,007 adverse, 511 null-band, and 134 right-censored rows. Those findings remain evidence about the two frozen Study A substitutes.

The next question is whether that adverse result persists when the controller is learned through full-sequence optimization, the network has twice as many nodes and four times as many allocated recurrent slots, one model enforces selected excitatory/inhibitory constraints, the source training worlds are separated from development and evaluation worlds, and condition labels remain hidden until raw scores are frozen. This is a replication question across implementations, not permission to average away the Study A result.

Study B implements two new 32-node learned rate networks without importing the Study A model or analytics code. The first permits mixed-sign recurrent columns. The second uses 26 excitatory and six inhibitory units, constrains every outgoing recurrent column to the sign assigned to its source unit, and assigns different leak constants to the two populations. Both use the same 1,024 allocated recurrent slots, the same exact 358-edge mask density, six inputs, one learned readout, matched training examples, matched optimization steps, matched checkpoint format, and the same blind evaluator. They are independent replacements for the Study A systems; they are not independent research teams, and they share one Study B evaluation implementation.

The Dale-constrained network is a **biologically constrained artificial substitute**, not a faithful biological reconstruction. It omits spikes, dendritic morphology, glia, neuromodulators, metabolism, gene expression, development, and evaluation-time synaptic plasticity. Its E/I ratio and sign constraint test whether selected biological structure changes the engineering conclusion. They do not make its hidden units neurons or its state a human mind.

16.2 Earned model definition

The ordinary problem is that a fixed hand-authored recurrent controller may conceal whether a result is caused by the capture ladder or by one convenient toy dynamics. A stronger test must first learn a stateful controller, preserve its training provenance, instantiate successors with declared subsets of its state, and then score those successors in worlds that training never accessed.

For both learned models, hidden state $h_t \in [-1, 1]^{32}$, input $x_t \in \mathbb{R}^6$, leak vector $\lambda \in (0, 1]^{32}$, recurrent matrix $W \in \mathbb{R}^{32\times 32}$, input matrix $U \in \mathbb{R}^{32\times 6}$, bias $b \in \mathbb{R}^{32}$, readout $r \in \mathbb{R}^{32}$, and body terms g_B, b_B obey

$$h_t = (1-\lambda) \odot h_{t-1} + \lambda \odot \tanh(W h_{t-1} + U(x_t \odot e) + b + 0.05\, s)\,, \tag{69}$$

$$p_t = \sigma\Big(g_B r^\top h_t + b_o + b_B\Big)\,, \qquad a_t = \mathbf{1}[p_t \geq 0.5], \tag{70}$$

where $e \in \mathbb{R}^6$ is a frozen input-gain vector and s is the centered mean of four explicit self-state fields. Units are dimensionless synthetic state, task step, and probability. These variables do not inherit milliseconds, volts, spikes per second, or biological energy units.

For the E/I model, the declared outgoing sign of source unit j is $q_j \in \{+1, -1\}$, with 26 positive and six negative sources. Projection after each gradient step enforces

$$W_{ij} = m_{ij} q_j |\theta_{ij}|, \qquad m_{ij} \in \{0,1\}, \qquad \sum_{ij} m_{ij} = 358. \tag{71}$$

The model is trained by deterministic, clipped full-sequence backpropagation through time on binary cross-entropy plus the frozen recurrent/input/readout penalty:

$$\mathcal{L} = -\frac{1}{NT} \sum_{n,t} [y_{nt} \log p_{nt} + (1-y_{nt}) \log(1-p_{nt})] + 10^{-4} \Big(\|W\|_F^2 + \|U\|_F^2 + \|r\|_2^2\Big)\,. \tag{72}$$

No early stopping is allowed. Each of 16 source models receives 48 epochs over the same twelve training worlds. C1 demonstration-only successors receive 16 imitation epochs over four separate development worlds. The 48 source runs and C1 reconstruction runs all end below their first recorded loss, with zero gradient-clipped epochs. This shows that the declared optimizer changed the learned parameters in the intended direction on its training objective. It does not prove task optimality or biological learning.

16.3 Frozen capacity, data access, and power

Eight source seeds, two architectures, five capture conditions, two reconstruction seeds, two body conditions, and four held-out worlds produce

$$N_{\text{blind jobs}} = 8 \times 2 \times 5 \times 2 \times 2 \times 4 = 1{,}280. \tag{73}$$

The build freezes 16 source checkpoints and 160 successor checkpoints before evaluation. The source uses twelve training-world seeds. C1 imitation uses four development-world seeds. All evaluation uses four different held-out-world seeds. The three sets are disjoint, and the training pipeline reports zero held-out-score generation and zero held-out-outcome access.

The source agent remains the inferential cluster. Under the preregistered planning values $\sigma_a = 0.10$, local margin $\delta = 0.15$, familywise $\alpha = 0.05$, four primary endpoints, one-sided $\alpha_m = 0.0125$, and target power 0.80, a normal approximation gives 4.224 source clusters before rounding. Rounding up and applying the 20% attrition allowance requires seven; Study B uses eight. This is development-replication power under assumed variance, not final confirmatory power and not an assertion that repeated worlds replace independent source agents.

16.4 Blinding and custody before interpretation

The trainer writes 1,280 opaque job files. The evaluator's manifest exposes only `blindId` and `jobRoute`. It does not expose architecture, source seed, capture condition, reconstruction seed, body condition, or world seed; its source code contains no blind-key route. The evaluator produces 1,280 score bundles, 19,200 metric rows, 61,440 translated internal-state rows, 15,360 paired intervention rows, and 1,280 failure rows while reporting zero blind-key reads and no aggregate outcome.

Only then are all six raw artifacts hashed into the raw-score freeze. The freeze records zero pre-freeze blind-key reads. The analysis refuses to proceed if any raw hash changes. Two post-freeze implementation failures are retained: one incorrect local seal route stopped before key access, and one parser rejected an intentionally absent normalized field after key access but before aggregate analysis. Neither failure changed a model, checkpoint, endpoint, score, margin, multiplicity rule, or raw hash. The completed analysis preserves this ledger rather than presenting a frictionless fictional history.

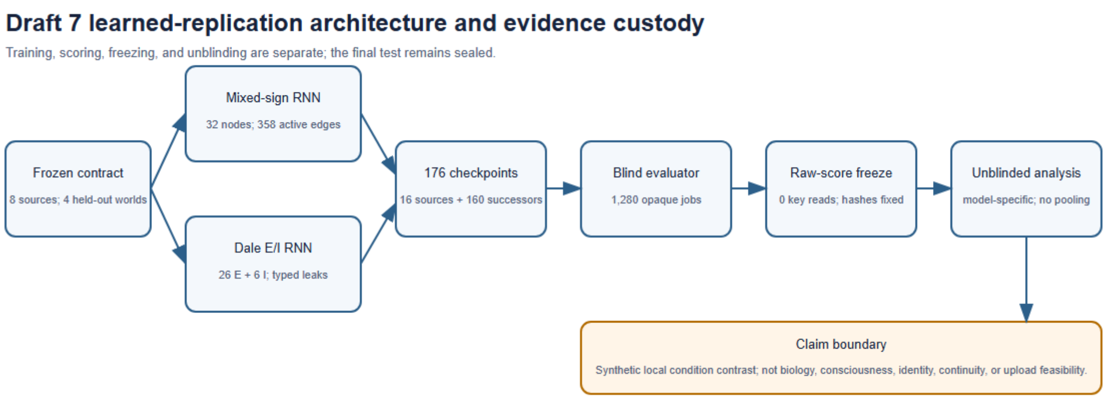

Figure 1: Study B architecture and evidence custody

16.5 Primary result: a local contrast, not an upload verdict

For source a, learned architecture k, and primary metric m, Study B retains the Study A estimand

$$\Delta_{akm} = \overline{Y}_{akm,C6} - \overline{Y}_{akm,C7}. \tag{74}$$

Four thousand fixed-seed bootstrap resamples operate on the eight source-level differences. Bonferroni correction fixes the one-sided alpha at 0.0125 per endpoint. The complete interval must lie inside $(-0.15, 0.15)$ for the label `local_equivalence`. This label means only that the measured C6-C7 difference is small under one model and endpoint. It is not equality of agents, state completeness, identity, continuity, consciousness, or biological equivalence.

Learned architecture	Metric	C6-C7 estimate	Frozen familywise interval	Local disposition
mixed-sign RNN	MU-M03 held-out functional performance	-0.002255	[-0.003966, -0.000640]	within local margin
mixed-sign RNN	MU-M05 intervention performance	-0.008243	[-0.009324, -0.007153]	within local margin
mixed-sign RNN	MU-M08 changed-body transfer	0.001751	[0.000784, 0.002932]	within local margin
mixed-sign RNN	MU-M13 adversarial pass fraction	-0.004557	[-0.008464, -0.000651]	within local margin
Dale E/I RNN	MU-M03 held-out functional performance	-0.000520	[-0.002688, 0.002411]	within local margin
Dale E/I RNN	MU-M05 intervention performance	-0.006042	[-0.007922, -0.004522]	within local margin
Dale E/I RNN	MU-M08 changed-body transfer	0.004383	[0.001343, 0.008511]	within local margin
Dale E/I RNN	MU-M13 adversarial pass fraction	0.000651	[0.000000, 0.001953]	within local margin

Thus Study B contains eight model-specific local-equivalence dispositions, zero non-equivalent dispositions, zero unresolved dispositions, and no cross-model disagreement on this one C6-C7 contrast. No pooled favorable label is issued. Model-specific estimates and difference-in-differences remain separate.

This result does **not** overturn Study A. It demonstrates that the C6-C7 conclusion is implementation-sensitive: the two Study A substitutes strongly separated C6 and C7, while the two learned Study B substitutes do not. That sensitivity is itself adverse to a universal claim. It means neither the favorable Study B contrast nor the unfavorable Study A contrast can be promoted into a model-independent conclusion about which state families are sufficient.

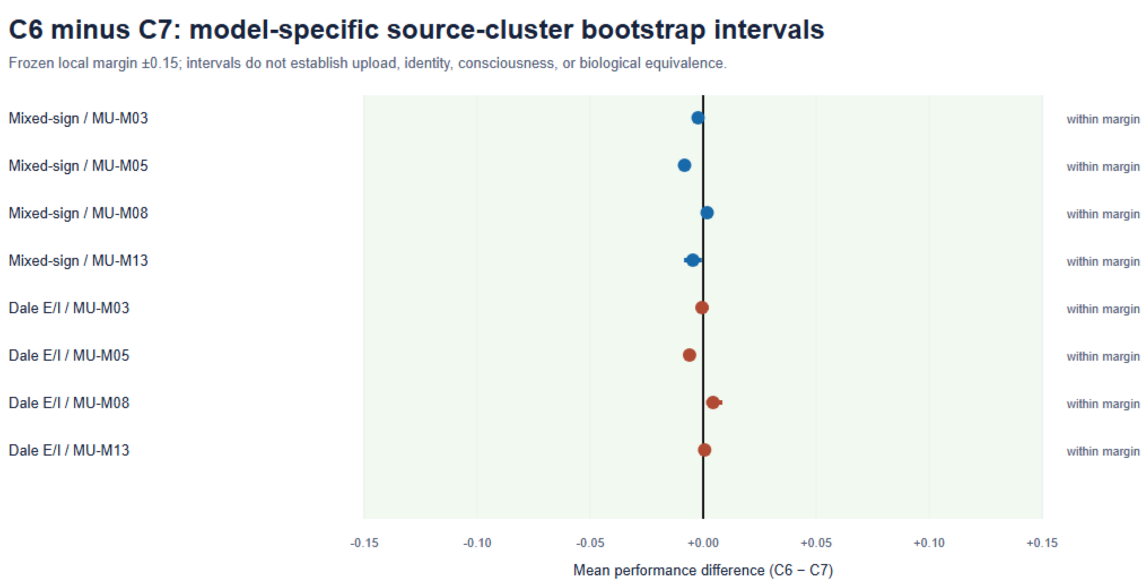

Figure 2: Model-specific C6-C7 intervals

16.6 Negative evidence remains visible

Study B retains all 19,200 metric rows: 4,131 adverse, 72 null-band, 412 right-censored, 868 descriptive, and 13,717 favorable. It also retains 4,156 intervention failures. The right-censored rows are time-to-action-divergence observations with no divergence inside the tested 24-step horizon; they are not imputed as known divergence times.

The local C6-C7 result coexists with important failure surfaces. Across capture conditions, MU-M13 adversarial performance is notably weaker for the mixed-sign C1 imitation condition (mean 0.2272) and remains variable in intermediate snapshot conditions. Body-readout remapping has the lowest aggregate intervention pass rate in both models: 0.5766 in the mixed-sign model and 0.6078 in the Dale-constrained model. No nonfinite state occurs, but bounded numerical stability does not convert an intervention failure into success.

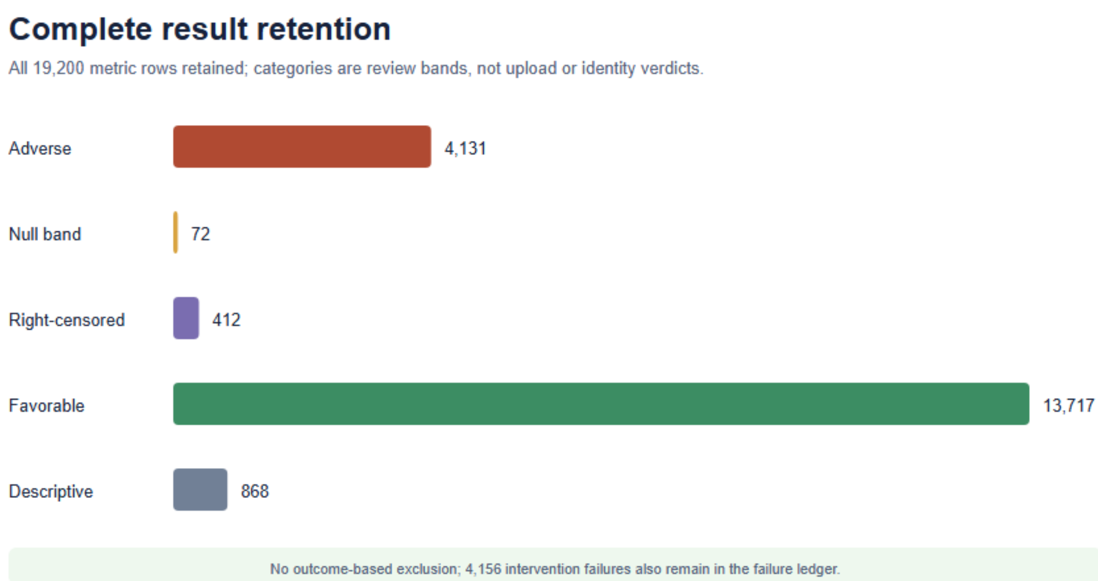

Figure 3: Complete result retention

16.7 Internal states, stability, and timescale boundary

Each source and successor contributes a translated 24-step trace containing action, state hash, mean, variance, norm, positive fraction, maximum absolute value, and sign pattern. Raw hidden vectors are intentionally absent from this release surface. The translation is a lossy audit representation and cannot establish semantic identity or phenomenology.

All 1,280 job pairs remain finite over 24 steps. The largest hidden-state magnitude is 0.729231 in the mixed-sign model and 0.681510 in the Dale E/I model. Maximum source/successor spectral radii are 0.356264 in the mixed-sign family and 0.559599 in the Dale E/I family. Those results bound the tested rollouts only. They do not prove asymptotic stability, arbitrary-input robustness, or a biological timescale correspondence.

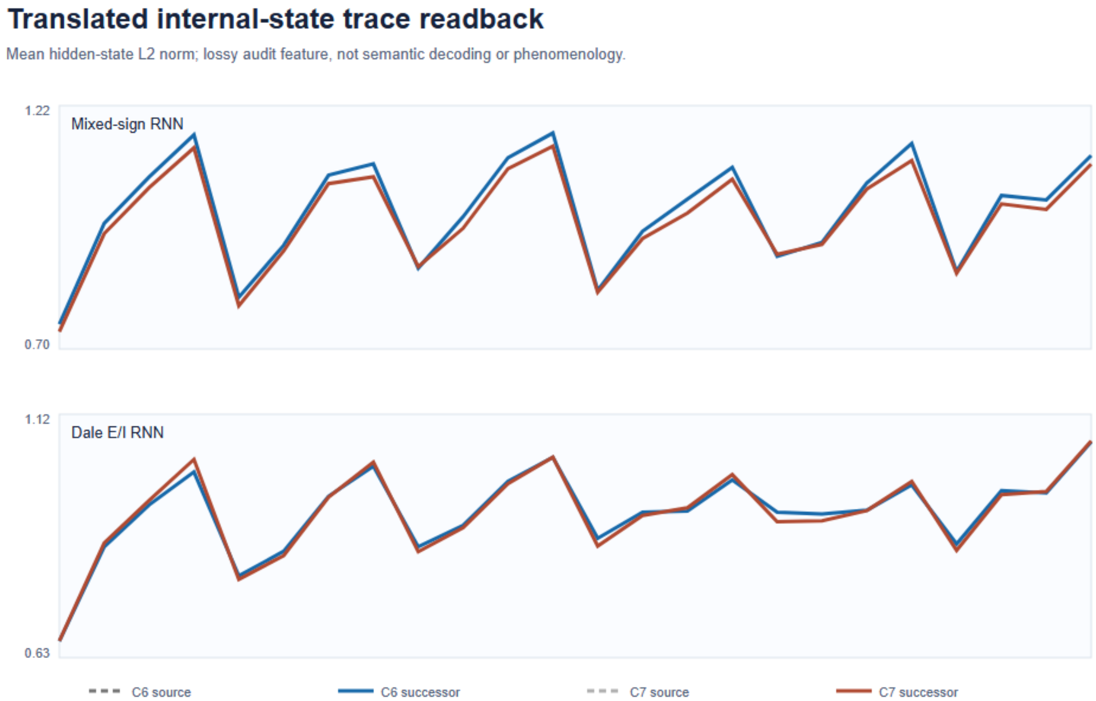

Figure 4: Translated internal-state traces

16.8 Uncertainty, variance, and the no-pooling rule

The balanced descriptive decomposition attributes variation to source, architecture, capture condition, reconstruction seed, body, world, model-by-capture, capture-by-body, capture-by-world, and an unassigned residual. Capture condition is the largest declared component for MU-M05 (0.5266). The residual is largest for MU-M03 (0.3759), MU-M08 (0.5999), and MU-M13 (0.4211). Capture-by-body is also material for MU-M03 (0.1562) and MU-M05 (0.1411). These fractions are descriptive sums of squares. They are neither causal variance components nor fitted random-effects estimates.

The Study A and Study B findings must therefore remain stratified by implementation. A convenient average across four models would destroy the exact fact that implementation changes the sign and disposition of the conclusion. The appropriate statement is: **local C6-C7 equivalence appears in both learned Study B substitutes and in neither Study A substitute; the cross-draft difference is unresolved model dependence, not evidence that uploading succeeded.**

16.9 Known implementation limit discovered by testing

The independent Study B test suite found a representation-level edge case. After the mixed-sign source recurrent matrix is serialized to 12 decimal places, reconstruction reapplies the row-norm projection. The resulting maximum recurrent-weight difference is approximately 1.000006×10^{-12} in C3, C5, C6, and C7. The Dale E/I model has zero observed difference under the same check. This numerical drift does not alter the four primary functional endpoints, but it can prevent an exact recurrent-family hash match and therefore affects exact-provenance metrics such as MU-M11. Study B retains the behavior and discloses it; it does not post-outcome repair checkpoints or recompute frozen scores. A later implementation should make canonicalization idempotent before preregistration.

16.10 Validation and exact replay

The frozen training stage produces 16 source and 160 successor checkpoints, 1,280 epoch rows, exact parameter replay for one frozen source seed in each architecture, and architecture checks on every epoch and checkpoint. The blind evaluator exactly replays its sentinel jobs, and the independent test suite recomputes the first, middle, and last blind score bundles from their opaque inputs.

The Study B application validator passes 60/60 checks. The independent test suite passes 37/37 tests. The visual readback verifies five SVG/PNG figure pairs. The package includes the preregistration, typed configuration, model and evaluator source, 176 checkpoints, opaque jobs, raw-score freeze, unblinded ledgers, intervention and failure data, translated internal-state traces, variance analysis, resource reconciliation, figures, tests, validation, and manifests.

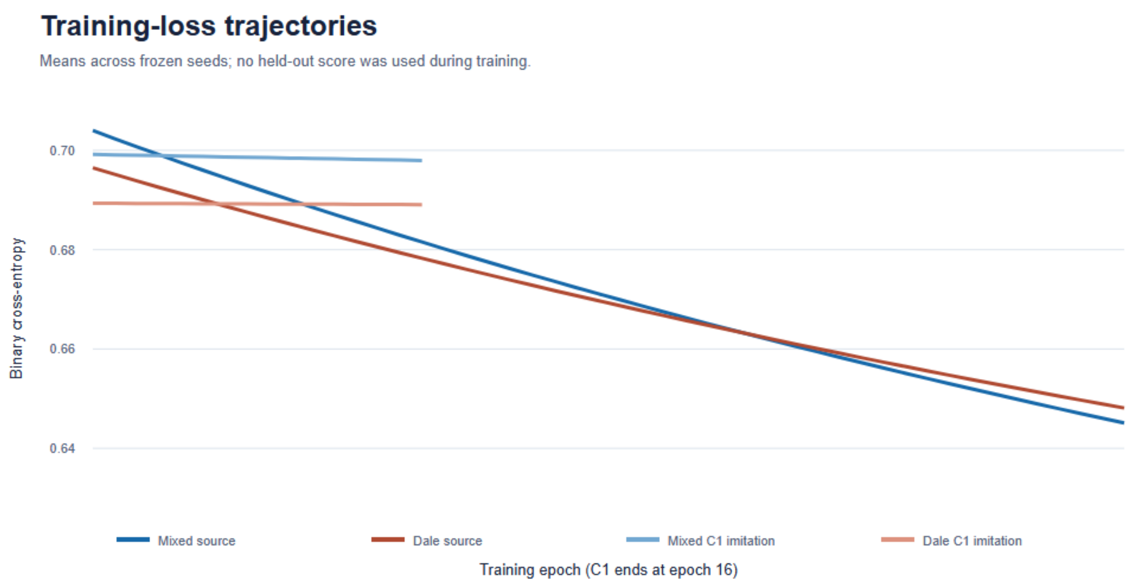

Figure 5: Training-loss trajectories

16.11 What the learned-model study establishes

Study B establishes that:

- an independently written learned-model pipeline can execute the declared source-training, capture, reconstruction, blinded held-out scoring, score-freezing, and unblinding sequence;
- two matched 32-node models, including one with selected Dale/EI constraints, produce the same local C6-C7 disposition on all four primary endpoints;
- the Study A adverse result remains intact and exposes implementation sensitivity rather than disappearing;
- checkpoint, translated-state, intervention, failure, and score provenance can be replayed under a frozen 1,280-job plan; and

- adverse, null, censored, and failed-intervention evidence can remain visible alongside a locally favorable contrast.

Study B does not establish a faithful brain, a complete state ontology, biological reconstruction, consciousness, numerical identity, personal continuity, human upload feasibility, long-horizon stability, or final-test performance. It also does not establish architecture independence: the contrast with Study A is direct evidence that implementation remains a live source of uncertainty.

The next honest gate is a frozen replication by a separately developed training and evaluation implementation, using more realistic learned embodied tasks, longer horizons, additional independently trained source agents, explicit spike- or conductance-based baselines if biological claims are contemplated, and evaluator separation enforced at the process or team level. The final-test plan remains byte-identical and `sealed_unopened`; Study B performs zero final-data reads, zero final-data writes, and zero unseals.

17. Bounded Formal Invariants for Evidence and Custody

17.1 The problem before the formal vocabulary

A benchmark can be numerically reproducible and still make an invalid logical jump. An interval can be mislabeled. Two models can disagree and then disappear inside a convenient average. A right-censored time can be treated as though the event time were observed. An omitted state family can be credited as exactly preserved. A revoked permission can be ignored. Two distinct successors can both be described as literally identical to one source. These failures are not repaired by collecting more synthetic measurements; they require rules that make the prohibited inference unavailable in the benchmark itself.

The Formal-Invariant Layer therefore adds a small machine-checked layer around the evidence ladder. It does not formalize a brain or decide what consciousness or personal identity is. It formalizes what the benchmark is allowed to say after receiving declared inputs. The reader can understand the operation without a coined term: classify one interval under one fixed margin, retain disagreement instead of averaging it away, keep censored observations typed as censored, require inclusion and a matching custody value before exact-preservation certification, block execution after revocation, and distinguish numerical successors under ordinary equality. These operations are called **bounded formal invariants** only after their work is explicit.

The formal carrier is finite and deliberately modest. Equivalence bounds use signed fixed-point integers. Their scale must be declared outside the theorem, and the margin, lower bound, and upper bound must share that scale. The application validator remains responsible for the scientific prerequisites $m > 0$, $\ell \leq u$, a valid estimator, a valid interval procedure, and a declared endpoint. Lean checks the decision logic after those values are supplied; it does not prove that an empirical interval is well estimated.

17.2 One interval receives exactly one local disposition

For fixed-point margin (m), lower bound (ell), and upper bound (u), the classifier is

$$D(m,\ell,u) = \begin{cases} \texttt{local\_equivalent}, & -m < \ell \ \wedge \ u < m, \\ \texttt{non\_equivalent}, & u \le -m \ \vee \ m \le \ell, \\ \texttt{unresolved}, & \text{otherwise.} \end{cases} \tag{75}$$

Lean proves that this function returns one of the three constructors, that each constructor has exactly the stated condition, and that the inside-margin and outside-margin conditions are incompatible for a positive margin and an ordered interval. This is a proof of the classifier's partition, not a proof that the Study A or Study B confidence intervals have biological meaning.

The strict interior matters. An interval that touches either margin is not certified as locally equivalent. A wholly separated interval is non-equivalent. Every remaining crossing or ambiguous interval stays unresolved. No fallback converts unresolved evidence into a favorable label.

17.3 Agreement is not sufficient for pooling

The ordinary problem with pooling is that identical labels can still come from dependent implementations. Study B's mixed-sign and Dale-constrained models share one training and evaluation package. Both return local C6-C7 dispositions, but their agreement does not establish an independent replication. Study A also gives a sharply different implementation-family result. The formal rule therefore requires two conditions before any pooled constructor exists:

$$P(i,d_1,d_2) = \begin{cases} \texttt{pooled}(d_1), & i = \texttt{true} \ \wedge \ d_1 = d_2, \\ \texttt{no\_pool}, & \text{otherwise,} \end{cases} \tag{76}$$

where i is an externally established independence certification, not a value inferred from matching outcomes. Lean proves both directions needed here: disagreement forces `noPool` regardless of i, and uncertified dependence forces `noPool` even when both labels are locally equivalent.

The exact primary summaries are also encoded as typed records:

$$R_6 = (0,5,3,1), \qquad R_7 = (8,0,0,0), \tag{77}$$

with fields ordered as local-equivalent, non-equivalent, unresolved, and model-disagreement counts. Lean proves $R_6 \neq R_7$ and that the cross-draft primary result cannot be pooled. The scientific interpretation remains the one already established in Study B: the difference exposes implementation sensitivity. It does not turn either result into an upload verdict.

17.4 Censoring, capture honesty, and provenance remain typed

The Formal-Invariant Layer represents a measured value, a right-censoring lower bound, and a missing value with different constructors. The only extraction function that returns an observed value returns `none` for right-censored and missing cases. This prevents a censoring bound from silently becoming an observed event time. It does not estimate a survival curve or choose a censoring model; those remain statistical responsibilities outside the finite theorem.

Exact preservation also receives an intentionally severe rule:

$$C_{\text{exact}} = I_{\text{included}} \wedge I_{\text{hash match}}. \tag{78}$$

Lean proves that an omitted state family cannot be certified as exact regardless of the second flag, and that an included but mismatched family is not exact. A hash match remains a custody statement about the declared digital object. It does not establish semantic, functional, biological, phenomenal, or personal identity.

Each memory event carries exactly one typed provenance constructor: inherited, acquired after reconstruction, inserted externally, or unresolved. `Unresolved` is not erased or treated as inherited. The theorem proves totality of the declared type; the application still must supply trustworthy event identifiers and evidence for its classification.

17.5 Permission, rollback, ladder separation, and fission

The permission model has `allowed` and `revoked` states. Revocation maps either prior state to `revoked`, and Lean proves that execution is then blocked. This is an application invariant, not a complete moral theory of consent. Real consent also requires authority, comprehension, scope, time, capacity, and a trustworthy interface.

Rollback certification is equality between the declared pre-intervention state identifier and the restored-state identifier. Lean proves that a known mismatch cannot be certified. This does not guarantee that an identifier captures every relevant physical or semantic state; it prevents the benchmark from calling a declared mismatch a successful rollback.

The evidence ladder is protected by a concrete countermodel. A `structuralOnly` record passes the structural rung and fails the functional rung. Lean therefore proves that there is no unconditional theorem from structural success to functional success across all declared ladder records. Any later bridge requires its own explicit axiom or theorem and cannot be smuggled in through the word *upload*.

Finally, for arbitrary objects under ordinary equality, Lean proves that if two successors are distinct, they cannot both be equal to one source. This is the narrow fission invariant. It does not select a philosophical theory of personal identity, and it does not deny qualitative similarity, causal inheritance, shared memories, or continuity claims under a different explicitly defined relation.

17.6 Machine-checking, typed proof piece, and axiom audit

The Formal-Invariant Layer package contains one Lean 4 source file, a natural-language statement, a typed proof-piece signature, compatibility edges, compiler output, an axiom audit, validators, and hashes. The source uses no `sorry`, `admit`, or user-declared `axiom`. Each promoted theorem is followed by `#print axioms`, so the compiler receipt records its dependency status rather than asking the reader to trust a prose assertion.

The proof-piece interface names what the leaf provides, what it requires, its finite software scope, its absence of any projection or biological interface, and its forbidden overclaims. Its compatibility edges are exact only toward the Paper 11 benchmark contract. Every proposed bridge to biological reconstruction, consciousness, numerical identity, personal continuity, or upload feasibility is absent or explicitly blocked.

The proof status is therefore split rather than inflated:

- **green in the stated finite application model:** the checked classifier, no-pooling, censoring, capture-honesty, provenance, revocation, rollback, ladder-countermodel, and ordinary-equality fission theorems;
- **blue only:** the Compact Development Pilot, Study A, and Study B synthetic execution results to which these decision rules can be applied;
- **gray or open:** biological interpretation, phenomenal consciousness, and personal-identity theory; and
- **not authorized:** any claim that the formal leaf proves mind uploading, consciousness equivalence, or human feasibility.

17.7 What the formal layer establishes

The Formal-Invariant Layer establishes that the declared benchmark can encode and machine-check a narrow set of inference and custody rules without weakening the adverse Study A result or pooling it with Study B. It turns several prose safeguards into typed constructions whose prohibited branches cannot be produced by the checked functions. It also preserves the complete internal manuscript lineage byte-for-byte, the 0/5/3/1 versus 8/0/0/0 implementation-family disagreement, all adverse/null/right-censored evidence, and the unopened final-test seal.

The Formal-Invariant Layer does not establish that the benchmark captures all relevant state, that the models are biological, that any successor is conscious, that a person continues, or that mind uploading is feasible. It does not settle the best statistical interval estimator, certify implementation independence, or prove the external inputs honest. The Strongest-Form Review therefore applies a lossless strongest-form challenge: an adversary must reconstruct the complete source-faithful argument before criticizing it, test every formal premise against the actual application contract, and retain rather than compress the Study A/Study B disagreement. The final test remains `sealed_unopened` with zero reads, writes, or unseals.

18. Strongest-Form Challenge

18.1 The problem before the red-team vocabulary

A critical review can fail in two opposite directions. It can accept a striking result because it likes the conclusion, or it can attack a shortened substitute for the argument because the substitute is easier to defeat. Neither operation tests this paper. A valid challenge must first recover what the paper actually says, including its genealogy, definitions, evidence ladder, experimental limits, negative results, and distinctions among structural resemblance, dynamical reproduction, functional performance, causal continuity, consciousness, and personal identity. Only then may the challenge identify a premise that is unsupported, an inference that outruns its evidence, or a test that cannot distinguish the proposed operation from an alternative.

The Strongest-Form Review therefore treats source-faithful reconstruction as part of criticism rather than as an optional courtesy. The rule is not that every claim must be accepted. The rule is that the object under review must be the strongest version the sources and manuscript actually support. An apparent mismatch triggers a search for omitted context, scale, timescale, receiver, endpoint, or translation before it can become a finding. If that search leaves an empirical burden unresolved, the burden stays open. It is not converted into a favorable result, but it is also not mislabeled as a contradiction between SAN and science.

For challenge (c), admissibility requires

$$A(c) = S(c) \wedge C(c) \wedge E(c) \wedge T(c) \wedge B(c) \wedge R(c), \tag{79}$$

where S is source recovery, C is full-context reconstruction, E is an exact external-evidence check, T is translation across terminology, scale, timescale, variable, receiver, and endpoint, B identifies the claim's actual burden of proof, and R preserves adverse, null, censored, failed, and model-dependent results. A challenge that fails any term is returned for repair rather than counted as an objection to the full argument.

The possible dispositions are deliberately nonbinary:

- **bounded by the existing claim:** the challenge attacks an overclaim that the manuscript already excludes;
- **answered by evidence or construction:** the declared benchmark supplies the relevant finite answer;
- **open empirical obligation:** the paper identifies a real test, but present evidence does not complete it;
- **open philosophical obligation:** the relevant identity or consciousness relation remains theory-dependent;
- **open governance obligation:** authorization, welfare, custody, or accountability cannot be reduced to a technical flag; and
- **implementation limitation:** the software result is real inside its model but does not support a wider generalization.

These dispositions do not hide negative evidence. They locate it at the claim class it can actually bear.

18.2 The argument reconstructed before challenge

The strongest source-faithful form of this paper's argument is not "scan a brain and copy a person." It is the following joined proposal.

1. A biological person is a changing, embodied, distributed process whose constraints include connectivity, morphology, cellular state, synaptic efficacy, ongoing activity, body coupling, memory history, goals, and recurrent interaction with a world.
2. Different reconstruction methods can preserve different subsets of those constraints. No one subset receives the name *mind* merely by being measurable.
3. Structural capture, state capture, dynamical reproduction, held-out functional performance, causal response, phenomenal evidence, and personal identity are different endpoints. Passing one cannot silently substitute for passing another.
4. Equivalence is receiver-, task-, body-, environment-, horizon-, metric-, and margin-relative unless a stronger theorem establishes invariance.
5. A copied successor can resemble a source while immediately diverging because its location,

evidence, body, action, and history differ. Branching makes numerical-identity claims especially demanding.

6. A scientific program can still make progress without deciding personal identity. It can state what was captured, test counterfactual responses, measure divergence, retain failures, and expose which state families matter for which endpoints.
7. The Upload Ladder Benchmark is an executable measurement architecture for that program. the executable studies instantiate artificial test cases; Formal-Invariant Layer checks narrow decision and custody rules. None is a human upload experiment.
8. The synthetic results are intentionally mixed. Study A returns 0 local-equivalent, 5 non-equivalent, 3 unresolved, and 1 disagreement/no-pooling outcome. Study B returns eight model-specific local-equivalence outcomes on one C6-C7 contrast, but no pooled favorable result. Their difference is evidence of implementation sensitivity.
9. Consciousness and numerical personal identity remain additional burdens. Behavioral and causal similarity can be important evidence without becoming an identity theorem.
10. Consent, revocation, rollback, memory provenance, welfare, and legal status remain part of the scientific design rather than an afterthought.

Every challenge in the Strongest-Form Review targets this ten-part argument. None is allowed to replace it with a connectome-only, behavior-only, substrate-independent, or already-feasible upload claim.

18.3 Thirty strongest-form challenges and their actual dispositions

ID	Strongest admissible challenge	Strongest-Form Review disposition
D9-C01	Several 2012-2013 source files have internal dates but later immutable Git fixation, so the earlier date cannot be presented as equivalent to contemporaneous public custody.	Source chronology remains split by date class. The concepts inform genealogy; only the later fixation supplies immutable Git custody unless a stronger public route is recovered.
D9-C02	The sentence "brain uploading can be done" could make the paper appear to claim feasibility.	Exact speaker recovery shows that sentence belongs to an interlocutor and Micah's answer begins "Prove it." The source-faithful contribution is the situated-copy and divergence challenge.
D9-C03	"The brain as a special kind of hard drive" could be read as a static storage metaphor.	The 2017 source describes a massively parallel, state-dependent, reciprocal read/write system. Literal static-disk compression is inadmissible; the operational metaphor remains bounded.
D9-C04	No present measurement captures every biologically relevant variable in a human brain.	Open empirical obligation. The paper requires declared endpoint-specific state sufficiency and never claims complete human-state observability.
D9-C05	A connectome omits conductances, receptor states, neuromodulation, intracellular conditions, and changing synaptic efficacy.	External neuroscience supports the objection to connectome sufficiency and the paper already makes those variables part of the typed state question. Their full measurement remains open.

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ID	Strongest admissible challenge	Strongest-Form Review disposition
D9-C06	The declared state model may still omit glial, vascular, immune, metabolic, extracellular-field, and molecular-history variables.	Open empirical obligation. Omission cannot receive exact-preservation credit; future tests must add a state family when it improves prediction or causal rescue.
D9-C07	Embodiment is not a static input-output shell; body and decoder can coadapt over time.	Supported by the BCI and body-remapping literature already cited. The synthetic body transforms test only a narrow engineering analogue, not human coadaptation.
D9-C08	Representational drift means identical function need not require identical microscopic activity.	The paper treats trajectory and function as separate endpoints. Drift motivates invariant and relational metrics; it does not license ignoring state when state matters causally.
D9-C09	Different hidden states can generate the same finite observations, making the source structurally unidentifiable.	Open identifiability obligation. Mathematical-Specification Stage supplies observability and countermodel tests, but no finite pilot proves a human state ontology identifiable.
D9-C10	Even an identifiable model may be practically unrecoverable under noise, destructive measurement, limited bandwidth, or unavailable interventions.	Open practical-identifiability obligation. The present application controls synthetic access and therefore cannot establish biological recoverability.
D9-C11	Finite held-out task equivalence permits many non-equivalent mechanisms and cannot establish general functional identity.	Correct and retained. The local label is task-, metric-, horizon-, body-, and world-specific; universal functional identity is not issued.
D9-C12	The 0.15 equivalence margin is a design choice, and another scientifically justified margin could change dispositions.	Open endpoint-calibration obligation. The strict classifier is valid for its supplied margin; external calibration and sensitivity analysis remain necessary.
D9-C13	Eight source clusters and an assumed standard deviation provide only a development-scale power calculation.	Implementation limitation. The power plan is explicit but not a population-level biological guarantee; independent sources and empirically estimated variance are required.
D9-C14	Multiple correlated endpoints can make favorable selection easy even with Bonferroni control on a designated family.	The primary family is frozen and corrected, while all other rows remain descriptive. Future confirmatory work must preregister endpoint families and dependence handling again.
D9-C15	A right-censoring lower bound cannot be treated as an observed divergence time, and naïve deletion can bias conclusions.	Formal-Invariant Layer blocks value extraction from censored rows. A declared survival model and censoring assumptions remain open statistical work.
D9-C16	Twenty-four- and thirty-two-step rollouts cannot establish long-horizon stability, identity, or retained memory.	Correct and prominent. Stability claims are limited to tested horizons; longer and multiscale horizons remain required.

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ID	Strongest admissible challenge	Strongest-Form Review disposition
D9-C17	Matching parameter slots does not guarantee matching expressive capacity, optimization difficulty, inductive bias, or information access.	Open model-comparison obligation. Slot and access matching prevent simple budget confounds but do not establish functional-capacity equivalence.
D9-C18	A Dale-sign constraint alone does not make an RNN biologically faithful.	Correct. Study B calls both models bounded substitutes and lists the missing biophysics. Spike-, conductance-, plasticity-, and body-constrained baselines remain future work.
D9-C19	Study B's two models share a generator, training family, evaluator, and research team, so their agreement is not independent replication.	Correct. No pooled result is issued; Formal-Invariant Layer proves that uncertified dependence blocks pooling.
D9-C20	Study A and Study B reach different primary dispositions, so any architecture-general conclusion is premature.	This is the central empirical challenge, not an inconvenience. Both exact summaries remain visible and unpooled; the program-level result is implementation sensitivity.
D9-C21	Thousands of adverse and failed-intervention rows could be obscured by reporting eight favorable local contrasts.	Strongest-Form Review rejects that compression. The 4,131 adverse, 72 null, 412 right-censored, and 4,156 failed-intervention Study B records remain prominent, as do Study A's adverse counts.
D9-C22	Exact hashes establish digital custody, not semantics, consciousness, or personhood.	Correct and machine-bounded. Hash claims remain object-level provenance only.
D9-C23	Translated state traces omit raw vectors and can miss semantically decisive differences.	Correct. The trace surface is a lossy audit view; it supports replay and diagnostics, not exact internal-state or phenomenal equivalence.
D9-C24	Lean checks functions over supplied finite inputs but cannot prove the inputs truthful, the estimator valid, or the state ontology complete.	Correct. Formal-Invariant Layer formalizes inference and custody logic only. Scientific input obligations stay explicit.
D9-C25	A no-`sorry` proof can still depend on standard axioms and can prove a statement too weak for the surrounding scientific claim.	Correct. The axiom inventory is disclosed theorem by theorem, and compatibility with upload, consciousness, or identity success is explicitly blocked.
D9-C26	Behavioral reports, self-model similarity, and causal response may still be insufficient to establish phenomenal consciousness.	Open empirical and philosophical obligation. The paper treats perturbation, rescue, and cross-theory indicators as evidence, not a decisive consciousness oracle.
D9-C27	Ordinary equality under fission does not settle psychological continuity, causal succession, narrative identity, or legal personhood.	Open philosophical and governance obligation. Formal-Invariant Layer proves only the narrow ordinary-equality fact.
D9-C28	Gradual replacement, scan-and-copy, external extension, and in-place repair may have different causal histories even when terminal functions match.	The ladder records route and continuity separately. No route is favored merely by intuition; route-sensitive experiments and explicit identity premises remain necessary.

Continued on next page

ID	Strongest admissible challenge	Strongest-Form Review disposition
D9-C29	A Boolean revocation state cannot exhaust consent, capacity, scope, welfare, or control over copies and memory edits.	Open governance obligation. The theorem blocks one prohibited software transition and is not presented as a complete ethics system.
D9-C30	A sealed final test and a complete synthetic package do not by themselves justify publication claims about biological uploading.	Correct. Strongest-Form Review leaves the final test sealed, issues no biological conclusion, and routes only the bounded framework and synthetic evidence toward later release review.

No challenge in the table authorizes deletion of an earlier adverse result. No bounded response turns an open burden into a completed experiment. The result is a sharper map of what the paper has actually established and what remains to be built.

18.4 State sufficiency is the central biological bottleneck

The strongest scientific challenge is not that a connectome has no value. Connectomes constrain possible routes, recurrent motifs, cell partners, and transformation paths. The challenge is that connection identity alone does not specify the operating state of the connected cells. Patch-seq, integrated morphoelectric classification, synaptic-strength measurements, neural activity atlases, and connectome-constrained modeling all point in the same direction: structure is informative, but predictive performance improves when the relevant cellular, dynamical, and body variables are included [8-18].

That evidence fits the paper's actual state architecture. It does not refute it, and it does not prove the architecture complete. The live question is endpoint-specific sufficiency. If a candidate state set (X_K) predicts or reconstructs an endpoint, the next test is not to declare (X_K) "the mind." It is to add or remove a state family (Z) under matched capacity and ask whether the conditional error changes:

$$\Delta_Z = \mathcal{L}(Y, \widehat{Y}_{X_K}) - \mathcal{L}(Y, \widehat{Y}_{X_K \cup Z}). \tag{80}$$

A reproducible positive Δ_Z under held-out intervention supports the relevance of Z to that endpoint. A small value supports dispensability only within the tested resolution, body, world, horizon, and model family. It cannot establish metaphysical irrelevance. This incremental design is how the state question becomes measurable without pretending that a complete human ontology already exists.

The same rule protects SAN's broader claim from a reverse strawman. The assertion that selfhood is distributed, embodied, recurrent, and history-dependent does not mean every microscopic variable must be copied with infinite precision. It means the burden is to discover which variables and relations remain causally necessary at each rung. Exact microphysical duplication, coarse functional emulation, and causal continuity are distinct hypotheses. The benchmark must let evidence distinguish them.

18.5 Identifiability and finite equivalence remain unresolved by good performance

The strongest computational challenge is underdetermination. A finite task suite may admit many models with indistinguishable outputs. A high score can therefore mean that one successor reproduced a relevant operation, or merely that the test did not probe the difference. The paper's observability, countermodel, changed-body, changed-world, intervention, and branch tests reduce that ambiguity but do not abolish it.

The Strongest-Form Review consequently preserves four separations.

First, structural identifiability asks whether distinct hidden systems can produce the same ideal observations. Second, practical identifiability asks whether finite noisy data can distinguish them. Third, task equivalence asks whether the systems behave similarly on a declared distribution. Fourth, causal equivalence asks whether matched interventions produce matched changes. None entails the next without additional premises.

The Study A and Study B disagreement is a direct demonstration of why these distinctions matter. Study A's bounded deterministic substitutes make C6 and C7 measurably different on the frozen primary endpoints. Study B's learned substitutes make the same contrast locally small. The result cannot be repaired by choosing the preferred architecture after seeing the outcome. Nor should Study A be allowed to dismiss the program, because both families are artificial substitutes. The correct inference is narrower and more useful: the state-sufficiency conclusion is not invariant across the tested implementation families.

An honest next experiment must therefore preregister at least one higher-capacity, separately implemented model family; expand source agents and environments; expose longer trajectories; and include mechanistically different baselines. If biological language is contemplated, spike- or conductance-based models, plasticity, neuromodulation, body dynamics, and explicit measurement mappings must enter the design. Agreement across those families would be stronger. Continued disagreement would also be informative because it would locate dependence on the assumed mechanism.

18.6 Statistical rigor does not convert a development model into a biological result

The Study B intervals are narrow because the synthetic generator is controlled, the task is bounded, and source-level differences are stable inside that model. That precision is real for the declared simulation. It is not automatically transportable to brains, patients, or arbitrary artificial agents.

Three burdens remain.

1. **Margin calibration.** The local margin of 0.15 must be justified for each endpoint by a meaningful consequence scale, not chosen because it yields a desired label. Sensitivity to plausible margins should be shown before a confirmatory run.
2. **Cluster uncertainty.** Eight source clusters exceed the frozen development calculation but remain too few for confident generalization to a heterogeneous biological population. Future work needs more independently generated sources and a variance model learned from relevant data.
3. **Censoring and horizon.** No observed divergence inside 24 steps is not infinite continuity. A survival analysis requires a declared censoring mechanism, longer follow-up, and a definition of

what event counts as divergence for each endpoint.

Multiplicity control also has a semantic dimension. Correcting p-values or interval coverage does not cure endpoint shopping if the endpoint itself is redefined after results are known. the synthetic studies freeze the primary family and retain every other row. Strongest-Form Review keeps that division: eight favorable C6-C7 intervals are a local primary result; thousands of adverse, null, censored, descriptive, and failed-intervention rows remain part of the total evidence surface.

18.7 Consciousness and identity cannot be smuggled through a functional metric

The strongest pro-upload argument may grant the state and measurement difficulties, then claim that sufficiently rich functional and causal equivalence is all that identity requires. That is a serious philosophical position, but it is a premise, not an experimental result. Other positions emphasize biological continuity, phenomenal organization, narrative continuity, or branching relations. The benchmark must state which relation is under test rather than letting the word *same* move between them.

This paper therefore preserves the branching case. If one source produces two distinct successors, ordinary numerical equality cannot identify both with the one source. That fact does not make the successors worthless or unreal. They may inherit memories, dispositions, projects, legal claims, and causal relations. It means only that a numerical-identity claim needs more than qualitative similarity.

Phenomenal consciousness has a parallel problem. A successor may report experience, preserve a self-model, generalize under changed conditions, and respond to perturbation and rescue. Those are valuable discriminators against simple scripts and disconnected lookup systems. They do not create a theory-neutral proof of experience. Strongest-Form Review retains cross-theory indicators and welfare precaution precisely because uncertainty remains. A system that plausibly supports suffering cannot be treated as disposable merely because philosophical certainty is unavailable; nor can precaution be rewritten as proof of consciousness.

18.8 The formal proof survives only at its declared interface

The strongest formal challenge concedes that every Lean theorem compiles, then asks whether the theorem is the scientific claim that matters. This is the right question. The Formal-Invariant Layer's answer is intentionally narrow.

The classifier theorem says that a fixed function returns the constructor corresponding to supplied fixed-point bounds. It does not construct a confidence interval. The pooling theorem says that a declared disagreement or uncertified dependence produces `noPool`. It does not certify independence. The censoring theorem blocks extraction of an observed value from a right-censored constructor. It does not fit a survival model. The custody theorem requires inclusion and a matching identifier. It does not prove that the identifier captures semantics. The revocation theorem blocks execution in a two-state machine. It does not prove informed consent. The fission theorem concerns ordinary equality. It does not select a personal-identity theory.

This separation is a strength only if it remains visible. A formal proof leaf becomes misleading when its name is wider than its statement or when its assumptions disappear in prose. The Formal-Invariant Layer package therefore exposes theorem names, types, axiom reports, compatibility edges, external obligations, and forbidden overclaims. The Strongest-Form Review confirms that the leaf fits the finite benchmark contract and has no valid edge to a claim of biological upload, consciousness replication, or personal continuity.

18.9 What survives the strongest-form review

The strongest-form challenge leaves four positive results intact.

First, the paper's historical argument survives source recovery: Micah's work repeatedly treats the self as distributed, changing, embodied, and relational; it challenges simplistic copy claims and develops reciprocal read/write and external-brain questions. The internal/public date distinctions and speaker correction remain mandatory.

Second, the typed evidence ladder survives. It is scientifically useful to prevent a wiring diagram, a state snapshot, a working simulation, a behavioral match, a causal match, a consciousness indicator, and a numerical-identity claim from collapsing into one word.

Third, the executable benchmark survives as a bounded research instrument. It can generate deterministic artificial sources, reconstruct them under declared capture conditions, test countermodels and interventions, retain provenance and failures, and report endpoint-specific uncertainty. Its synthetic nature remains part of the result, not a footnote.

Fourth, the mixed empirical record survives unchanged. Study A and Study B disagree. The eight Study B local intervals are real inside their learned substitutes. The adverse Study A dispositions, the Study B adverse/null/censored rows, and the intervention failures are equally real. No pooled or biological verdict follows.

The review also leaves seven open gates: biological state sufficiency; measurement feasibility; structural and practical identifiability; architecture-independent replication; long-horizon and multiscale stability; consciousness evidence; and personal-identity plus governance theory. These are the paper's research program, not defects to conceal.

18.10 Strongest-form result and release boundary

The Strongest-Form Review completes its lossless analysis without deleting one byte from the internal manuscript lineage, changing a model, weakening a criterion, selecting a favorable subset, or opening the final test. Thirty challenges were reconstructed against the actual argument. Challenges that expose real uncertainty remain open; challenges that target claims the paper does not make are bounded by the existing scope; and implementation limits remain attached to the synthetic results they constrain.

The strongest defensible conclusion is now exact: **this paper supplies a source-grounded, typed, executable, and partly machine-checked framework for asking what a reconstruction preserved; its synthetic studies demonstrate both measurable local similarity and strong implementation sensitivity; it does not establish biological mind uploading, replicated consciousness, numerical personal identity, or human feasibility.**

Any final-preprint release of this work must preserve the manuscript and reproducibility package and may proceed only if it preserves that sentence's full meaning, every unfavorable result, the complete source chronology, the no-pooling rule, the theorem boundary, and the `sealed_unopened` final-test state. Formatting, diagrams, and release metadata cannot promote a lower evidence rung into a higher one. No provider or publication action is authorized by the Strongest-Form Review itself.

19. Data, Code, and Reproducibility

The provider-ready release package includes the public manuscript, PDF, claim-source matrix, strongest-form challenge register, formal-proof receipts, validation records, and a companion reproducibility archive containing the deterministic reference implementation and the complete synthetic studies. The archive preserves configurations, code, tests, manifests, checkpoints, score ledgers, adverse/null/censored results, intervention failures, translated state traces, and the warning-free Lean compiler output.

The biological final-test plan remains sealed and unopened. The release archive includes only its public plan and custody seal, not final-test data. All reported executions are synthetic engineering studies. No component of the package establishes biological reconstruction, consciousness replication, numerical personal identity, personal continuity, or human mind-upload feasibility.

20. Author Contributions, Funding, and Competing Interests

Micah Blumberg conceived the framework, developed the Self-Aware Networks genealogy and Upload Ladder Benchmark, directed the computational and formal studies, interpreted the results, and wrote the manuscript. No external funding is declared. The author declares no competing financial interest.

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